Long-case viva session notes — Case Cards

M.S. Ophthalmology practical examination · Tamil Nadu Dr M.G.R. Medical University · 16 October 2026

Every card follows the same headings. The generic long-case proforma is in 00 · Index and master case format.

★ kept at JEH last year · Long case · Short case · Task · Chart

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VN1 Long-case viva session — at a glance

From the Cochin Ophthalmic Club live long-case session, 7 Mar 2021 · auto-captions

What this session is and how to use this guide

- **What it is:** a 4 h 05 min live mock exam. Seven postgraduate candidates presented long cases to senior examiners, who interrupted, criticised and asked viva questions. Closing advice from the examiners and a short glaucoma talk followed. The moderator called it a “survival kit”: it teaches the technique of answering and how to take constructive criticism, not only knowledge. The planned format was a 5-minute presentation, about 20 minutes of viva and 5 minutes of criticism.
- **What is in these five cards:** VN1 (this card) is the overview and timed summary. VN2 holds every survival tip, de-duplicated, with timestamps. VN3 covers the seven cases, what was criticised and the preferred diagnosis line. VN4 is the viva question bank with the answers the examiners expected. VN5 lists the slides and images mentioned and a plan for your last week.
- **How to use it:** read VN2 and VN3 first, then cover the answers in VN4 and test yourself. The session answers are what the examiners said on the day, not a textbook. Glaucoma answers were compared with your glaucoma cards, and every difference is flagged for you to settle with your seniors. Use the timestamps to jump to the exact moment on the recording.

Timed summary of the whole session

Timestamps are minutes and seconds, or hours, minutes and seconds after one hour. Each row is one real segment of the recording. The YouTube chapter titles are unreliable, so the rows follow the speech instead.

Time	Segment	What happened
00:12–02:08	Welcome and introduction	The host opens the monthly Sunday session as a mock exam in which candidates will not be grilled. Two guest examiners are welcomed. A glaucoma talk is promised for the end because World Glaucoma Week starts that day.
02:11–04:21	Format and examiners	The moderator explains the survival-kit idea and the 5 + 20 + 5 minute format. He introduces the two examiners by qualification and asks the first candidate to share her screen.
04:24–12:22	Case 1 presentation	A 60-year-old woman with bilateral primary angle-closure glaucoma (PACG) and cataract is presented in full. It takes 8 minutes against the 5 allotted.
12:25–25:08	Case 1 criticism by the first examiner	The history was fragmented and the complaint was in medical words. There was no optical or drug history and no differential. He goes through the negative history item by item and asks why each was asked.
25:08–28:28	Case 1 criticism by the second examiner	Each step should narrow the differential, and the history must not be back-fitted to the diagnosis. Avoid negatives you cannot defend, because what you say steers the questions.
28:28–37:14	Case 1 viva	Questions cover assessing a shallow anterior chamber, the treatment plan, the missing visual field report and the target intraocular pressure (IOP). The examiners prefer plain cataract extraction first.
37:14–39:05	Hand-over and screen-sharing	The next candidate struggles to share her slides in full-screen mode.
39:08–46:57	Case 2 presentation	A 67-year-old man with left central retinal vein occlusion (CRVO), iris neovascularisation and neovascular glaucoma is presented.
47:01–47:34	Investigations and discussion	The investigation list is cut short. The group moves on to discussion, since a presentation normally ends with the clinical examination.
47:34–50:39	Risk factors and refraction	The candidate must categorise the risk factors, name the likely cause and name the refractive error.
50:39–53:40	What to look for in a CRVO eye	The candidate gives acuity, pupil, iris and angle. The examiner adds lids, globe, conjunctiva and anterior chamber cells and flare.
53:40–56:50	Why iris new vessels and not disc new vessels	The candidate explains ischaemia and vascular endothelial growth factor (VEGF). She cannot explain why the iris, not the disc, is affected, so the examiner supplies the reason.
56:50–58:56	CRVO chronology	The time-line of the history is muddled and must be restated with exact durations. A hint leads to raised pressure and corneal oedema as the reason for the recent fall in vision.
58:56–1:00:33	Gonioscopy	How to tell appositional from synechial closure: do indentation gonioscopy and say so.
1:00:34–1:02:18	Classification and site of block	The candidate classifies vein occlusions and gives the site of block. The commonest quadrant for branch vein occlusion is asked, with two reasons.

Time	Segment	What happened
1:02:19–1:05:30	Is it ischaemic? The disc and hypertensive changes	The disc in vein occlusion, collaterals versus new vessels, artery versus vein on the picture, and Gunn sign.
1:05:30–1:06:56	Describing the fundus	The order of description is praised. Haemorrhage types are probed, and the label “ischaemic” is challenged because there are no cotton-wool spots.
1:06:57–1:10:30	Management of neovascular glaucoma	Prognosis is reframed around what the patient wants. The pressure-lowering drugs, drugs to avoid, atropine and steroid drops, and the type of glaucoma are discussed.
1:10:31–1:14:40	Retinal management	There is no true retinal neovascularisation, so the disc vessels are collaterals. Anti-VEGF with generic name and dose, macular oedema on optical coherence tomography (OCT), and steroid options with pressure monitoring.
1:14:41–1:16:30	Panretinal photocoagulation (PRP) and closing view	PRP is discussed. The second examiner thinks the eye was probably non-ischaemic first and became ischaemic later.
1:16:28–1:17:28	Hand-over to the proptosis case	A clock check, then advice to keep the history brief and write “nothing contributory” for the systemic examination.
1:17:29–1:22:51	Case 3 presentation	A 34-year-old man with painless axial proptosis of the right eye and disc oedema (compressive optic neuropathy) is presented.
1:23:14–1:25:35	Signs of proptosis and thyroid eye disease	Naffziger method, worm’s-eye view, exophthalmometers and the signs of thyroid eye disease. Möbius sign is added by the examiner.
1:25:36–1:28:10	Why it is not thyroid orbitopathy	No redness, no lid retraction and full movements. Retropulsion wording is corrected. Commonest causes of proptosis by age and sex.
1:28:11–1:31:00	Investigations	Blood tests, thyroid profile, computed tomography (CT) or magnetic resonance imaging (MRI), and orbital B-scan. Causes of axial proptosis, including vascular lesions and optic nerve tumours.
1:31:00–1:33:35	Minimal proptosis and exophthalmometry	Ruler test, absolute versus relative exophthalmometry, Luedde’s instrument. The examiner disagrees with thyroid disease as the diagnosis.
1:33:35–1:36:25	CT findings	Sharing the scan fails repeatedly, so the candidate keeps talking. The scan shows a globular, homogeneous intraconal mass.
1:36:25–1:38:44	Other causes and duration	Inflammatory and parasitic causes, myositis, a list of intraconal masses, and why the lump was noticed only in the last year.
1:38:44–1:41:20	Optic nerve compression and pseudoproptosis	Fields, colour vision (bedside desaturation test), pseudoproptosis and the pseudo-Graefe sign.
1:41:24–1:43:44	First examiner: general feedback	Be prepared, relaxed and confident. Use simple English, lay words for symptoms, and lead the examiner to topics you know.
1:43:46–1:47:13	Second examiner: preparation advice	Case templates, positives only at the bedside, saying “pardon” instead of guessing, weekly case presentation, and the examiners test basics.
1:47:13–1:48:14	Interlude	Viewing numbers are read out. No audience questions. The next candidate is asked to share her screen.
1:48:14–1:58:30	Case 4 presentation	A 73-year-old man with a left third nerve palsy: isolated, pupil-sparing, partial, probably microvascular.
1:58:31–2:03:15	Critique of the presentation	The purpose of the fundus examination, grading of ptosis, the type of ptosis and the wording of one slide.
2:03:18–2:08:40	Syndromes and localisation	If the palsy were not isolated: nuclear, fascicular, subarachnoid, cavernous and orbital sites, with four brainstem syndromes.
2:08:42–2:12:50	Lesion site, partial-palsy rules, cavernous sinus	The dictum about when a palsy cannot be called partial. The candidate argues against orbital and cavernous sinus lesions and refuses a bait of “nuclear lesion”.
2:12:34–2:17:27	Pupil sparing versus involving, and emergencies	Why the pupil is spared in ischaemia, what it means for management, and the red flags that make a third nerve palsy an emergency.
2:17:27–2:17:49	Hand-over	The next candidate begins.
2:17:49–2:24:41	Case 5 presentation	A 55-year-old woman with a perforated right corneal ulcer after a negative scraping.
2:24:41–2:27:50	Is it a suitable long case? Aetiological versus morphological diagnosis	A perforated ulcer is not a usual long case. The examiner reads the picture as probably peripheral ulcerative keratitis (PUK) that has perforated.
2:27:52–2:29:00	Bacterial versus fungal keratitis	The candidate lists the differences. No correction is given.

Time	Segment	What happened
2:29:03–2:32:44	Critique of the corneal drawing	Cross-sections at three areas, lids, lashes and tear film, a precise description, keratomycosis kept in mind, and no digital tonometry.
2:32:44–2:34:10	Adnexa and quality of the case	The adnexa means more than the lacrimal sac. The differential includes PUK, Mooren's ulcer, vasculitis and fungal or viral keratitis. The case is judged not clear-cut.
2:34:10–2:39:52	Management questions	Work-up for fungal keratitis, scraping site, B-scan, antibiotic choice, monotherapy, and Mooren's ulcer versus other PUK.
2:39:52–2:40:37	Syringing advice	A third examiner says to syringe in all infectious keratitis.
2:40:37–2:44:30	Management of PUK or keratomycosis	Control the infection before steroids, signs of control, peritomy for Mooren's ulcer, and the goal of treatment.
2:44:30–2:47:00	Large perforation	No glue for a large perforation with iris prolapse; therapeutic keratoplasty is preferred.
2:47:00–2:48:08	Recent investigations	Anterior segment OCT for impending perforation. Exams ask for applications of recent investigations.
2:48:08–2:48:35	Hand-over	The next candidate begins.
2:48:35–2:55:05	Case 6 presentation	A 49-year-old man presented as "uveitis". Right eye red and painful for a week, with recurrent episodes and high myopia.
2:55:06–2:56:57	Summary and diagnosis	The candidate gives a multi-part diagnosis with cataract first and uveitis last.
2:57:03–3:01:50	First examiner's questions	Laser in situ keratomileusis (LASIK) in only one eye, how pseudoproptosis was diagnosed, axial length, how to find LASIK evidence, and the unexplained movement limitation.
3:01:52–3:09:50	Second examiner revises the diagnosis	The primary diagnosis is bilateral axial, possibly pathological, myopia. The cataract is complicated, not senile. Refractive surgery must be checked by keratometry. Uveitis is a lower-order diagnosis.
3:09:51–3:10:47	Wrap-up	Usually a short case and sophisticated for a postgraduate, but well done. The take-home point is to tailor and present any case properly.
3:10:47–3:19:13	Case 7 presentation	A 44-year-old man with sudden painless loss of vision in the left eye: pseudophakic retinal detachment (RD) after earlier vitreoretinal surgery.
3:19:13–3:24:08	First examiner questions the RD case	Missing slides, the detail of the injury, scars not sought, three risk factors, and the hunt for the break.
3:24:08–3:35:42	Second examiner's viva on RD	Slit-lamp signs, Shaffer sign, Lincoff's rule, degenerations that need treatment, management, then pathological myopia and refractive options in high myopes.
3:35:42–3:38:46	Closing remarks on the RD case	"Seek the hole and seal the hole", external versus internal procedures, and how to keep searching for the break.
3:38:46–3:43:28	Closing advice from a senior male examiner	Start from day one, treat patients as your books, document exactly, carry a notebook and pocket books, and re-read the basics.
3:43:30–3:50:06	Closing advice from the female examiner	Hard work, a year-by-year plan, discussion groups, patients as the encyclopedia, and managing stress. The seven candidates are congratulated.
3:50:06–3:51:00	Thanks and viewing statistics	The live stream has about 934 views.
3:51:01–3:53:16	World Glaucoma Week announcements	The club's programmes for the week and the sponsor are announced.
3:53:16–4:03:44	Talk: medical management of glaucoma	A practical talk by a senior glaucoma specialist on pressure as the key risk factor, tonometry, drug classes, newer agents, preservatives, drug allergy and choosing drops.
4:03:58–4:05:45	Closing	The organiser thanks everyone. The recording is cut off.

Limits of the source

- Only the auto-generated captions were available. There were no slides and no video frames, so everything said about pictures comes from the speakers' words.
- The captions garble medical terms. Obvious errors were corrected from context, and a [?] marks every reading that is not certain.
- An audio dropout at 07:05–08:25 hides part of the first case's anterior segment description. The speaker is unclear in several exchanges, especially in the third and fourth parts.

- Where an examiner gave a figure or a rule, it is reported as stated in the session and not checked against a textbook. Only the glaucoma answers were compared with your glaucoma cards.
- Names of candidates, examiners and patients are left out on purpose.

Read more: YouTube: Ophthalmology PG Long case Live — Survival tips explained (Cochin Ophthalmic Club), https://www.youtube.com/watch?v=e3X1W9wrC_A

VN2 Survival tips — what the examiners told candidates

From the Cochin Ophthalmic Club live long-case session, 7 Mar 2021 · auto-captions

Every distinct tip, grouped by the stage of the exam

Repeated tips from the four parts of the recording are merged into one line, and a merged tip keeps all its timestamps. Some tips were given in words and some are clear from how the examiners behaved; the second kind is marked "(implied)". Numbering restarts in each group. Where a tip uses a technical term, the matching question in VN4 gives the examiners' answer.

Before you start

1. The moderator framed the session as a survival kit: it teaches the technique of answering and how to take constructive criticism, not only knowledge. [02:23–02:34]
2. Start preparing from day one, not on the eve of the exam. There is no replacement for hard work, practice and perfection. [3:39:10, 3:43:39–3:43:47]
3. Know which phase of training you are in. Ophthalmology needs about three months before you start to love it, because undergraduate exposure is small. [3:43:56–3:44:20]
4. In the first six months acquire every basic clinical skill and instrument technique. Categorise yourself, learn the terminology and the newer procedures, and master the equipment (gonioscopy, slit lamp, indirect ophthalmoscope, 90 dioptre lens) and the tests. [3:39:30–3:39:38, 3:47:27–3:47:46]
5. In the first six months write down every new term each day and read about it for half an hour every evening. [3:44:20–3:44:37]
6. Do one refraction every day and read up retinoscopy and refraction, so that you are proficient by six months. [3:47:46–3:47:56]
7. Read up every investigation that comes with your cases. [3:47:56–3:48:03]
8. In the second six months pick up clinical signs, make diagnoses and miss nothing. In the second year learn to advise patients, plan investigations and surgery, and master cataract surgery and its complications. In the third year miss nothing and discuss the differentials of each case. [3:48:03–3:48:45]
9. Choose one textbook that suits you and add your own simplified, categorised notes (sticky notes) to it, to revise in the final month. [3:44:37–3:45:15]
10. Treat every case as your Bible and every patient as your encyclopedia. The more you see, the more you learn, so do not complain about workload. [3:39:24, 3:45:17–3:45:48]
11. After each patient ask yourself the take-home message. [3:46:00–3:46:10]
12. Document every case fully and exactly. Write the real findings every day, not notes copied forward from the day before. [3:39:53–3:40:18]
13. Carry a small notebook, record the cases you see and discuss them with seniors. Call patients back after a couple of days and re-examine them once you have read up. [3:40:24–3:40:42]
14. Do not use a phone or the internet as a substitute for knowledge. Build your own memory. [3:40:44–3:41:15]
15. Re-read the same basic clinical books again and again (two of the books named were Parsons and Kanski; the third title is garbled). Revisit each day's cases and ask the faculty about every doubt. [3:41:17–3:41:35]
16. Carry a pocket clinical-methods book (a second-edition "Clinical Methods" text was recommended [?]) and a second recommended book (title garbled [?]) every day. Look things up in front of the patient or in the next room. [3:41:56–3:42:40]
17. Keep colour pencils, a circle template (a bangle works) and a transparent scale in your pocket or purse, so that you do not run around when a ptosis or similar case is given. Also carry a basic armamentarium: a tuning fork, and a stethoscope and hammer [?]. [3:42:42–3:43:13]
18. Protect your study time. Treat the three years (about a thousand days) as the time to study and skip worldly distractions. Cope with stress by loving your work, and do not use family or children as an excuse for not studying; make them your strength. [3:46:13–3:46:45, 3:47:02–3:47:20]
19. Present cases as early and as often as possible, at least once or twice a week, so that exam day is not the first time. Practise oral presentation a lot, because fluency comes out automatically under exam tension. [3:39:41–3:39:53, 1:45:59–1:46:19, 3:48:45–3:48:58]
20. Form small discussion groups and discuss cases and signs as often as you can. This reduces the fear of scary cases. [3:49:00–3:49:25]
21. Ophthalmology books are not reference books. Understand the natural course of each disease (for example how a retinal detachment behaves) and answer from understanding, not from lookup. [3:49:33–3:49:56]
22. Revise the basics that examiners keep asking. These are synechial closure versus iris processes, and Gunn sign with the arteriovenous-crossing changes of hypertensive retinopathy. Also learn the two reasons why branch retinal vein occlusion is commonest in the superotemporal quadrant: the branching pattern with a direct thrust, and the many arteriovenous crossings. [1:00:28–1:00:31, 1:01:48–1:02:13, 1:05:20–1:05:30]

23. Keep revising high-yield topics such as third nerve palsy, and make notes of all the points, because it has many dictums. One example: an isolated medial rectus palsy or an isolated mydriasis is not called a partial third nerve palsy. [2:08:58–2:09:36, 2:12:15–2:12:25]
24. Be proficient in uveitis basics: the telltale signs of old uveitis, the signs of recurrent anterior uveitis, old versus fresh keratic precipitates, and the types of keratic precipitates. [3:09:19–3:09:41]
25. Learn to tailor and present whatever case you are given, and never miss the importance of any finding. [3:09:41–3:09:51]
26. Know exactly which peripheral retinal degenerations need treatment and which do not. It is a very common question, because every postgraduate examines high myopes with the indirect ophthalmoscope. Be ready to explain to the patient which lesions need intervention. [3:28:40–3:29:29]
27. Check your slides before you present. Put in every key picture (a primary-position photograph and the anterior-segment pictures were missing in two cases), and know how to enlarge and share them in full-screen mode. Examiners will ask to see them. [37:33–39:00, 1:48:50–1:48:58, 3:19:16–3:19:40]
28. While a colleague presents, listen critically for mistakes. The examiner invited the other postgraduates to spot the error (implied). [1:58:57–1:59:03]
29. Expect ordinary cases. A perforated ulcer is not a normal long case. Keratitis is usually a short case, not perforated, often healing and often fungal, because acute patients should not be inconvenienced. [2:24:48–2:25:25]

Presenting the history

1. Give exact durations in days, weeks or months, and keep the time-line consistent. Do not write “sudden” and “progressive” without saying when each applied. [56:56–57:45]
2. Describe the history fully, especially the decrease of vision. Do not give a fragmented, telegraphic history. [12:56–13:09]
3. Present the presenting complaint in the patient’s own words, for example “coloured rings around the light”, and not in medical terms such as “rainbow halos”. Use lay language for symptoms and keep sign-words for the examination, because sign-words in the history sound tutored. [13:43–14:09, 1:42:14–1:42:25]
4. Give the optical (spectacle) history under its own heading, and always include a drug history. Link every past-history question to management: steroid exposure, and drugs that are contraindicated in asthma or cardiac disease. [14:16–14:38, 18:09–18:13, 24:51–25:08]
5. Form a provisional diagnosis or differential from the history alone before you move to the examination. For gradual loss of vision in both eyes in an elderly person, run through all the causes (lens, glaucoma, retina, systemic and diabetic, cornea) before you settle. [14:52–14:59, 15:38–16:10]
6. Let each history point and each examination step narrow the differential, so that the next question follows from the last answer. Do not tailor the history backwards to the final diagnosis. Present it as a prospective history in which the diagnosis emerges. [25:24–26:14, 26:40–26:50]
7. Every question you ask must serve a purpose. Know what each negative history item is ruling out. [21:19–21:22, 21:36–21:51]
8. Leave out negative history that does not belong to the case. If you ask about seizures because of headache, you should also be asking about other neurological deficits. [23:25–23:29, 27:11–27:21]
9. Be ready to defend anything you say or write. Ask yourself “why have I said this?” [27:21–27:33]
10. State clearly whether a past procedure such as laser in situ keratomileusis (LASIK) was done in one eye or both. A vague history is the first thing examiners probe. [2:57:07–2:57:15]
11. Patients call any corneal refractive procedure “LASIK”, so do not accept the label without evidence. Cross-check the history, for example by asking whether the patient saw well without glasses after the surgery. Records are rarely available, so examination is how you corroborate the history. [2:59:44–2:59:54, 3:06:33–3:06:54]
12. After trauma, elaborate the mechanism of injury (what the patient was doing) and try to get the patient’s records. [3:19:55–3:20:14, 3:20:29]
13. Reason from the time-line. Normal vision for 10 years after a penetrating injury argues that the entry was not in the cornea or lens but further back, probably in the sclera. [3:20:45–3:21:05]
14. In orbital inflammation ask about pets and parasite exposure. [1:36:25–1:36:40]

Examination and findings

1. Be ready to say what you looked for in each part of the examination, and tie it to the findings already presented. For example, with a blood pressure of 220/110, look first for hypertensive changes in the fundus, not only for papilloedema. [1:59:15–2:00:20]
2. Know the textbook grading of signs cold, for example mild, moderate and severe ptosis with the millimetre values. [2:00:35–2:00:55]
3. Label the neurological examination “neurological evaluation” and give a one-line gross summary (no sensory-motor deficit, deep tendon reflexes grossly normal). Do not use an odd heading such as “muscular skeletal system”. [2:02:55–2:03:15]
4. State the lens grading, for example the grade of nuclear sclerosis. [32:20–32:23]
5. When you say the lids and globe are normal, know what you are excluding, for example xanthelasma and proptosis. Mention the relevant adnexal and conjunctival signs. [51:57–52:26]
6. In a vascular occlusion look for cells and flare in the anterior chamber, to pick up inflammatory causes. [53:12–53:36]

7. Link the presenting event to a mechanism. If the intraocular pressure (IOP) is high, check the cornea for epithelial oedema and bullae and say what you looked for. [58:56–59:10]
8. In gonioscopy, say that you did indentation gonioscopy and state whether the closure was appositional or synechial. [59:46–1:00:20]
9. In any vein occlusion, examine the disc and know why: risk factors, glaucoma, and collaterals versus new vessels. [1:03:00–1:03:15]
10. Be able to describe artery versus vein and point them out on a picture. [1:04:56–1:05:00]
11. Describe the fundus in an order: disc in detail, macula, vessels, then background. Do not jump to the pathology and skip the rest. [1:05:35–1:05:54]
12. Know how to describe each type of haemorrhage and a microaneurysm, how to tell a dot haemorrhage from a microaneurysm, and which conditions give red lesions in the fundus. [1:06:21–1:06:50]
13. Say “resistance to retropulsion” rather than just “positive” or “negative”. [1:26:35–1:26:44]
14. Know the clinical signs of thyroid eye disease. The examiner has a list of about 28 and named Möbius sign (difficulty in convergence) as one of the earliest. [1:25:11–1:25:36]
15. Know a simple bedside test for minimal proptosis, the ruler test. [1:31:15]
16. Know several exophthalmometers: Hertel, Luedde, and one for use after an orbitotomy. [1:32:36]
17. When you suspect optic nerve compression, test colour desaturation with a red or green object, not only with plates. [1:39:56–1:40:24]
18. In the corneal diagram, draw cross-sections at three areas: the oedematous stroma, the apparently normal cornea and the site of perforation. [2:29:13–2:29:35]
19. Draw the lids with lashes and the tear film. Draw a normal tear meniscus as a triangle at the lower lid margin, and a dry-eye meniscus as concave with small marks for debris. Blue with hatching for stromal oedema was approved. Examiners say this gives the impression that you examined well. [2:30:39–2:31:35]
20. Describe a corneal lesion cornea first, then the lesion, precisely and not vaguely (“two-thirds of the cornea”). Concentrate on the cornea and mention the pre-corneal tear film, lids and lashes. [2:29:40–2:30:12]
21. In a corneal ulcer examine and mention the adnexa, not only the lacrimal sac: lid margin, pre-corneal tear film, Meibomian glands, lash and lid-margin position, and pre-existing dry eye. In the other eye mention the tear meniscus, and think of dry eye when the picture looks like peripheral ulcerative keratitis. [2:30:04–2:30:24, 2:32:46–2:33:17]
22. Do not perform or mention digital tonometry in a perforated cornea. It can make the condition worse. [2:32:23–2:32:41]
23. If you blame an orbital inflammatory cause for limited movements, examine for it. Evert the upper lid, retract it laterally and ask the patient to look down. Then look at the lacrimal gland for injection or tenderness. [3:01:23–3:01:40]
24. When a patient reports LASIK, look for a flap scar. You cannot see it unless there is gross thinning, so press on the cornea with a blunt instrument or a cotton-tip applicator (after topical anaesthetic [?]) to bring out a scar. After photorefractive keratectomy (PRK) there may be nothing to see. [2:59:56–3:00:30]
25. In a centre without topography, use keratometry to check for earlier refractive surgery: a flat reading, or trouble getting a reading, suggests it, and a keratometry of 44–45 suggests none. Reason from the finding: minus 12 dioptres with flat keratometry means regression, partial correction or no correction, and the diagnosis must be revised. [3:05:21–3:06:30, 3:06:59–3:07:17]
26. Use exact descriptive terms, for example circumcorneal congestion and not just “hyperaemia”. [3:07:25–3:07:47]
27. When you have few findings, elaborate fully on the ones you have; “tessellated” alone was not enough. In a myopic fundus describe the disc size and colour, cup–disc ratio, tilt, vessels, peripapillary atrophy, chorioretinal degeneration at the peripapillary region and macula, and posterior staphyloma. Examiners want the basics described well so that they can take you forward. [3:03:56–3:04:04, 3:08:19–3:09:15]
28. After any trauma or surgery, look around the limbus and over the conjunctiva and sclera for scars and report them. After a foreign-body history, a posterior removal leaves a scar, possibly a blackish discolouration. [3:21:12–3:21:22, 3:37:25–3:37:55]
29. Identify the type of cataract surgery from the scar and state it definitely. Say the lens is in the bag if you can see the capsulorhexis rim in front of the lens. [3:21:36–3:21:57, 3:22:04–3:22:08]
30. When you cannot see a break, still say where it is likely to be (the laser-mark area, or a pigmented patch from the foreign-body entry). Use indentation indirect ophthalmoscopy and search again and again; take a break and search again. [3:22:56–3:23:45, 3:38:03–3:38:23]
31. Pigment or cells in the anterior vitreous are a Shaffer sign only in an eye that has not been operated. In an operated eye they may reflect the intervention. [3:26:39–3:27:08]
32. For IOP measurement, seat the patient comfortably, use the correct technique, calibrate the tonometer and disinfect it. [3:55:29–3:55:42]

The diagnosis line

1. Re-read your diagnosis slide before you present. The examiners go straight to it and check every grade and finding written there. [1:58:49–1:59:10]
2. Be specific in every term: “senile cataract”, not “cataract”. State the type that applies to your patient, for example “neurogenic ptosis”, and not a general list. [15:15–15:20, 2:02:08–2:02:19]

3. Make the label match what you can show. An “ischaemic” label with no cotton-wool spots will be noticed. [1:05:56–1:06:07]
4. Be able to name the refractive error when asked, not only quote the numbers, even though refraction is a separate case in the exam. [50:06–50:21]
5. State clearly why a leading diagnosis is unlikely. For thyroid orbitopathy, the points were no inflammation, no lid retraction and full movements. [1:25:36–1:26:11]
6. Be ready to give both an aetiological diagnosis and a morphological diagnosis. [2:25:27–2:25:39]
7. When the picture is unclear, keep a spread of differentials in mind: peripheral ulcerative keratitis (PUK), Mooren’s ulcer, vasculitis-related immune keratitis, and infection by fungus or virus. [2:33:19–2:33:40]
8. Be ready for “what if it were not isolated?” and have the associated syndromes ready. [2:03:21–2:03:35]
9. Never use a label such as pseudoproptosis unless you can define it. Proptosis is forward protrusion of the globe in primary gaze; pseudoproptosis is due to a shallow orbit or a large globe. Exclude orbital and facial causes and measure the axial length. [2:58:09–2:58:44]
10. Rank the diagnosis as primary, secondary and tertiary. Being told “it is a uveitis case” does not make uveitis the primary diagnosis. Even when tipped off that “this is a cataract case”, place each diagnosis by its importance. [3:02:30–3:02:50, 3:07:50–3:08:14]
11. Carry your own measurements into the diagnosis. A 30 mm globe in both eyes means axial myopia. [3:03:31–3:03:39]
12. Avoid “senile” for a 49-year-old (the cut-off quoted was about 45). With high myopia call the cataract a complicated cataract, that is, a cataract with an associated morbidity. [3:04:26–3:05:00]
13. Name the factors that predispose an operated eye to detachment (vitrectomy, cataract surgery and capsulotomy with the neodymium:yttrium–aluminium–garnet (YAG) laser). Always list the associated risk factors for rhegmatogenous retinal detachment: YAG capsulotomy, trauma, surgery and high myopia. [3:22:25–3:22:40, 3:25:36–3:26:04]

Investigations and management

1. Bring the relevant investigation reports with a glaucoma case, such as the visual field and glaucoma imaging. A glaucoma diagnosis needs a documented field defect. [34:22–35:11]
2. Learn how to interpret a visual field report, because most examiners ask it. The examiner recommended a journal article on presenting a visual field report. [35:13–35:32]
3. Know the target IOP and the precautions during surgery. [35:45–36:10]
4. A stepwise plan is acceptable: control the pressure medically first, then plain cataract surgery, then check the IOP, add medicines and do glaucoma surgery later if needed. Frame it as a first aim with several options depending on the response. [33:51–34:13, 36:56–37:10]
5. Tailor the investigations to rule out the risk factors that have not already been detected. [49:43–49:53]
6. Back a claim of macular oedema with the confirming test, optical coherence tomography (OCT), and its numbers (central thickness above about 270–280 microns), plus the fundus sign of a lost macular reflex. [1:12:57–1:13:30]
7. Tie the prognosis and plan to what the patient wants, which is better vision, and not only to the laboratory diagnosis. [1:07:14–1:07:24]
8. Say that the systemic evaluation is done and controlled, and move to ocular management when you are told. [1:07:34–1:07:42]
9. Use generic drug names with the correct dose, and no brand names. [1:11:57–1:12:10]
10. When you use steroids, state that you will monitor the IOP. [1:14:34–1:14:44]
11. Computed tomography (CT) is often enough for proptosis, even when vision is reduced. Do not insist on magnetic resonance imaging (MRI) unless it is needed. [1:28:50–1:29:30]
12. Know the red flags that make a third nerve palsy an emergency: pupil involvement, acute onset, severe headache, meningeal signs and hemiparesis. They mean immediate neurosurgical referral, and recognising them can save a life. [2:15:51–2:17:15]
13. Scrape the ulcer margin and base, remove necrotic debris and avoid thinned areas to prevent perforation. Do a smear straight away and look for fungal hyphae; if none is found, start empirical treatment guided by severity. [2:35:36–2:36:05, 2:35:41–2:35:49]
14. Order an ultrasound (B-scan) of the posterior segment when you cannot see in, especially if surgery is likely, to define the extent of infection. [2:36:36–2:36:57]
15. Antibiotic monotherapy is “not indicated” in general. The examiner allowed it only where fortified drops are not accessible and the lesion is mild or moderate, off the visual axis and responding. [2:37:25–2:37:57]
16. Do syringing in all cases of infectious keratitis; do not rely on the test of regurgitation on pressure over the lacrimal sac (ROPLAS) alone. Treating a lacrimal block helps the ulcer heal. [2:39:55–2:40:30]
17. Control the infection first and only then think of steroids, even in PUK. [2:41:30–2:41:36]
18. The goal of treating infectious keratitis is to eliminate the infection and prevent complications. Restoring vision is not the goal at this stage. [2:43:44–2:44:26]
19. For a large perforation do not offer glue. Plan therapeutic keratoplasty as early as possible once the infection is under treatment. [2:45:34–2:45:50]
20. Brush up on the applications of recent investigations, such as anterior segment OCT for corneal thinning and OCT in posterior segment cases. Examiners ask about them. [2:47:15–2:48:03]

21. Keep “seek the hole and seal the hole” at the centre of every retinal detachment answer. [3:36:07–3:36:12]
22. Classify detachment procedures in a simple way: external (scleral buckles) versus internal (silicone oil and gases). Know everything about the gases. [3:36:16–3:36:47]
23. In a vitrectomised eye a buckle is preferable but behaves differently. The final choice is the surgeon’s preference. [3:38:30–3:38:42]
24. When several drops are suspected in a drug allergy, stop all of them and reintroduce them one by one. [4:01:21–4:01:25]
25. For the first-line drug choice in glaucoma, use a prostaglandin analogue if affordable and a beta-blocker if cost is an issue. Use carbonic anhydrase inhibitors only for mild-to-moderate lowering. Add a second drug if the first works, and switch class if the first response is poor. [4:01:43–4:02:11]
26. Treat the person, not the disease. Glaucoma is chronic, so build a good rapport. [4:03:16–4:03:26]

Behaviour in the viva

1. Stay relaxed, cool and confident. If the examiner catches you out, stay calm. Smile, hide the tension, and answer with good articulation. Examiners praised a candidate despite visible tension and told the candidate to relax. [1:41:55–1:41:58, 1:43:24–1:43:41, 2:05:09, 2:06:07, 2:58:04]
2. Examiners differ, so do not be rattled by a harsh style. If you prepare every day, presenting is no problem: the presentation stays the same and only the examiners change. [1:43:15–1:43:24, 3:43:15–3:43:26]
3. Use simple English with scientific terms. Flamboyant English is not needed. [1:42:02–1:42:09]
4. “The ball is in your court”: examiners do not come with a fixed list, so your presentation leads them. Lead them to the areas you know well, and mention only what you can answer questions on. Avoid negatives you cannot defend, because what you say steers the questions, and do not tie yourself in a knot. [25:08–28:12, 1:42:27–1:42:45, 1:45:51–1:45:57]
5. If you do not know, do not guess badly. The examiner will see your gap and keep pressing. Say “pardon, sir” or “pardon, madam” and pause; never guess. [1:42:55–1:43:12, 1:45:34–1:45:43]
6. Never try to outsmart the examiner. [1:45:43–1:45:49]
7. Answer exactly what is asked. The examiner said “not answering my question” when a related detail was given instead. [1:45:47–1:45:51, 2:43:52–2:43:58]
8. When asked about your case, discuss your case and not an assumed scenario. Answer for this patient. The examiner cut in twice with “pertaining to this particular case only”. [32:02–32:10, 2:02:13, 2:45:00–2:45:04]
9. Do not “disappear” when you are asked a question. Answer it. [3:35:56]
10. Do not agree with a suggested diagnosis that your own findings contradict. The candidate said no to “nuclear lesion?” and the examiner approved (implied). [2:12:09–2:12:20]
11. Examiners catch one glaring mistake and then forget everything else you said, so double-check your terms before you use them. [2:57:52–2:58:02]
12. The examiner’s questions come straight from the standard text (“I am not asking anything out of the book”). Answer from the standard definition. [2:01:12–2:01:20]
13. Structure your lists. Categorise risk factors (systemic vascular, thrombophilia, inflammatory, blood dyscrasias) rather than listing them at random. Localise a lesion in a fixed anatomical order, here from the orbit to the nucleus, which is easier to follow. [48:19–48:27, 2:05:19–2:06:07]
14. Depth beyond what a postgraduate is expected to know is not demanded. When a candidate stalled on a retina-specialist “why”, the examiner supplied the answer and said so. [55:41, 56:37–56:44]
15. If a picture or slide will not load, keep talking and keep answering. [1:34:29]
16. The exam tests the basics. Missing a rare question will not fail you, but a basic blunder will, for example writing a retinoscopy reading of plus 4 and calling it myopia. [1:46:34–1:46:51]
17. The examiners test the flow of your reasoning and how you interpret the findings, not only the final diagnosis. [1:46:56–1:47:06]
18. Many examiners are subspecialists and not general ophthalmologists, so you may be better read than they are in some areas. Be ready for basic questions even from examiners outside the case’s subspecialty [?]. [1:46:19–1:46:34, 3:24:14–3:24:44]

Time management

1. Keep to time. The planned format was a 5-minute presentation, about 20 minutes of viva and 5 minutes of criticism, 30 minutes in all. A presentation of 8 minutes was flagged as too long, and each case was later cut to about 15 minutes in total. [02:11–04:21, 12:22, 25:08–28:28]
2. Keep the history brief. Leave out normal items and write “nothing contributory” for the systemic examination. [1:17:11–1:17:28]
3. The long-case presentation normally finishes with the clinical examination, and the discussion then takes over. [47:12–47:17]
4. Prepare a template for each common case type (proptosis, uveitis, open-angle glaucoma, neuro-ophthalmology and so on) with the examination columns ready. At the bedside write only the positive findings, and fill in the normal findings later, outside the room. [1:43:52–1:44:20, 1:44:22–1:44:44]
5. In a real-patient exam you have time, and you may be asked to elicit signs in front of the patient. Stay relaxed. [1:45:24–1:45:29]

The commonest mistakes the examiners criticised

- A fragmented history with the complaint in medical words (“rainbow halos”) and a muddled time-line without exact durations.
- Negative history items that the candidate could not defend (seizures, diplopia), a history back-fitted to the diagnosis, and no optical or drug history.
- No differential voiced from the history, and vague terms: “cataract” for “senile cataract”, “hyperaemia” for “circumcorneal congestion”, “senile” for a 49-year-old.
- A diagnosis list in the wrong order, with the presenting problem first (uveitis, or cataract) when the primary diagnosis was axial myopia. Also a label that did not match the findings (“ischaemic” with no cotton-wool spots).
- Labels used without proof or definition: pseudoproptosis with no axial length, LASIK accepted from history without keratometry, “partial” palsy without the rules.
- Basic grading and definitions not known by heart: ptosis grades, Gunn sign, retropulsion called “positive”, haemorrhage types.
- Gonioscopy reported without saying indentation was done, and no statement of appositional versus synechial closure.
- A thin fundus description (“tessellated” only), jumping to the pathology, or a generic purpose for the fundus examination (papilloedema when the blood pressure was 220/110).
- Missing material: no visual field report with a glaucoma case, no primary-position photograph, no anterior-segment pictures, no scars looked for after trauma or surgery, and no search for the break.
- Unsafe or wrong management: digital tonometry in a perforated cornea, glue for a large perforation, steroids started before the infection was controlled, no syringing, antibiotic monotherapy, and no pressure monitoring with steroids.
- Answering a different question, drifting to assumed cases, guessing instead of saying “pardon”, or going quiet when asked.
- Overrunning the time: 8 minutes against 5 for the presentation.

Read more: YouTube: Ophthalmology PG Long case Live — Survival tips explained (Cochin Ophthalmic Club), https://www.youtube.com/watch?v=e3X1W9wrC_A

VN3 The seven cases — what was presented, what was criticised, the preferred diagnosis line

From the Cochin Ophthalmic Club live long-case session, 7 Mar 2021 · auto-captions

The seven cases in the order they were presented

Each case below gives the patient as presented (age, sex and complaint only), the candidate's diagnosis, the examiners' preferred diagnosis line and the criticisms with their timestamps. Where the examiners dictated or restated a line, it is shown in a box you can read aloud. Where they gave only pieces, the pieces are listed as bullets. Case numbers follow the order of the recording.

Case 1 — Primary angle-closure glaucoma with cataract, both eyes [04:24–37:14]

- **Patient:** a 60-year-old woman. She has had decreasing vision in both eyes for 3 years, with intermittent left-sided eye pain and headache for 2 years.
- **Candidate's diagnosis [11:59–12:13]:** both eyes primary angle-closure glaucoma (PACG); both eyes compound hypermetropic astigmatism, against the rule; both eyes immature senile cataract; type 2 diabetes mellitus; systemic hypertension.
- **Examiners' preferred line:** no model line was dictated. The second examiner restated the diagnosis as below and asked the candidate to assume nuclear sclerosis grade 2–3 [31:08, 32:15–32:27]. The first examiner reminded the candidate that a glaucoma diagnosis needs a documented field defect [34:44–35:07].

> SAY IT
> Bilateral primary angle closure glaucoma with senile cataract.

What was criticised

- The history was fragmented. The decrease of vision should have been described more fully and clearly. [12:56–13:09]
- The complaint was reworded into medical terms ("rainbow halos"). It should have been in the patient's words, such as "coloured rings around the light". [13:43–14:09]
- There was no separate optical (spectacle) history, which is essential in a long case. Add the optical history and the drug history. [14:16–14:38, 18:09–18:13, 25:04–25:08]
- No differential diagnosis was voiced from the history. Reach a provisional conclusion before you start the examination. [14:41–14:59]
- The term "cataract" was too vague. Say "senile cataract". [15:15–15:20]
- The negative history items could not be defended: redness, diplopia, nausea, vomiting and photophobia, seizures, and trauma. Seizures were called not relevant. [21:36–24:00, 23:25–23:27]
- The past history lacked a drug history. Asthma and cardiac disease matter for steroid-induced cataract and for contraindicated drugs. [24:03–25:00]
- The history was not driven by differentials. The examiner said it had "gone retrospective", tailored to PACG. [25:12–26:14, 26:24–26:50]
- The candidate listed negatives unrelated to the case. [27:11–27:21]
- The lens grading was not given. [32:20–32:23]
- No visual field report, or other glaucoma report, was shown with the case. A glaucoma diagnosis should be backed by a documented field defect. [34:22–35:20]
- The presentation took 8 minutes against the 5 allotted. [12:22]
- Two inconsistencies were not raised by the examiners, but check your own case for the same. The complaint said 3 years while the history of present illness said one year. Closure was described as more than 180° in the summary but more than 270° in the viva. [04:24–12:22, 32:59–33:05]

On your cards: G4 (primary angle-closure disease), with G9 (gonioscopy), G10 (Humphrey visual field), G8 (applanation tonometry) and GX (treatment toolkit).

Case 2 — Left central retinal vein occlusion with iris neovascularisation and neovascular glaucoma [39:08–1:16:30]

- **Patient:** a 67-year-old man. He has had defective vision in the left eye for one week, on top of a sudden loss 8 months earlier that was never worked up.
- **Candidate's diagnosis [46:40–46:50]:** both eyes senile immature cataract; left eye ischaemic central retinal vein occlusion (CRVO) with macular oedema and neovascular glaucoma. In the discussion the candidate called the glaucoma "secondary closed-angle glaucoma due to neovascularisation", and the examiner accepted that wording [1:10:15].
- **Examiners' preferred line:** an examiner restated the diagnosis without correcting it [47:41–47:52]. The second examiner felt the eye was probably non-ischaemic at the start and became ischaemic with time, which explains the few cotton-wool spots [1:16:08]. The vessels on the disc were judged to be collaterals, not new vessels [1:11:37].

> **SAY IT**
> Left eye central retinal vein occlusion with neovascularisation of the iris and neovascular glaucoma. The glaucoma is secondary closed-angle glaucoma due to neovascularisation.

What was criticised

- The refractive error was given only as numbers. It should be named when asked. [49:57–50:21]
- The lids, globe, conjunctiva and anterior chamber findings relevant to systemic causes were not volunteered, so the examiner prompted for them. [51:30–53:36]
- No retinal neovascularisation was mentioned, and the candidate could not explain why the iris is involved without new vessels on the disc or elsewhere. [54:35–56:44]
- The history time-line was muddled: "sudden onset 8 months", then "progressive", then "a few days". Durations were vague. [56:56–57:45]
- The candidate did not volunteer the corneal and pressure-related reasons for the sudden drop in vision. [58:00–59:05]
- The gonioscopy report was incomplete. Say that indentation gonioscopy was done and whether the closure was appositional or synechial; the candidate wavered. [59:34–1:00:25]
- The candidate could not define Gunn sign, and knew the arteriovenous-crossing changes poorly. [1:05:08–1:05:20]
- The candidate could not at first show artery and vein on the picture. [1:04:44–1:04:58]
- The candidate was slow to say why the disc matters in a vein occlusion (risk factors, collaterals versus new vessels). [1:03:00–1:04:30]
- Haemorrhage descriptions lacked the basics (flame, dot, blot, microaneurysm), and the "ischaemic" label did not match the few cotton-wool spots. [1:05:56–1:06:50]
- The candidate gave no clear generic name with a dose for the anti-vascular endothelial growth factor (anti-VEGF) injection. [1:11:57–1:12:20]
- The investigation list was cut short before discussion began. This is not a criticism, only how the long case ran. [46:55–47:17]
- Praise: the presentation was "very nice and vivid" [47:34–47:36], and the sequence of the fundus description (disc, macula, vessels, background) was approved [1:05:32–1:05:45].

On your cards: G5 (neovascular glaucoma), G7 (CRVO) and G9 (gonioscopy with indentation). The retinal side of CRVO is not yet built — RETINA.

Case 3 — Axial proptosis of the right eye with disc oedema [1:17:29–1:41:20]

- **Patient:** a 34-year-old man. He has had painless, gradual forward protrusion of the right eye for 1 year and a mild dull ache behind the eye for 3 months.
- **Candidate's diagnosis [1:22:00–1:22:49]:** axial proptosis with compressive optic neuropathy. The first differential was thyroid orbitopathy, then intraconal masses (cavernous haemangioma, optic nerve meningioma, optic nerve glioma or schwannoma [?], circumscribed lymphangioma, orbital lymphoma).
- **Examiners' preferred line:** no model line was dictated. The examiner did not accept thyroid disease [1:25:36–1:26:11, 1:33:06]. The picture was an intraconal mass causing axial proptosis in a young man, with normal vision, no postural variation and no inflammation. The scan showed a globular, homogeneous intraconal mass [1:35:44–1:37:32]. Lymphoma was thought unlikely [1:38:00].

What was criticised

- The history should have been briefer. Skip the normals and write "nothing contributory" for the systemic examination. [1:17:11–1:17:28]
- Retropulsion was called "positive" and defined as the globe going in easily. The correct wording is resistance to retropulsion. [1:26:11–1:26:44]
- No thyroid eye sign was seen, and the examiner added Möbius sign (difficulty in convergence) as an early sign to look for. [1:25:11–1:25:36]
- The candidate had not asked about pets or parasite exposure. [1:36:25–1:36:34]
- Thyroid was the natural first thought, but the history and examination did not fit it: no inflammation, no lid retraction, axial proptosis and full movements. [1:25:36–1:26:11, 1:27:23–1:27:41]
- The candidate preferred magnetic resonance imaging (MRI), but the examiner said computed tomography (CT) was enough. [1:28:50–1:29:30]
- The candidate did not know the simple Luedde exophthalmometer. [1:32:08–1:32:35]
- For colour vision the candidate named a chart-based test (probably Ishihara [?]). The examiner wanted colour desaturation with a red or green object. [1:39:24–1:40:24]
- The CT would not share on screen. The examiner told the candidate to keep answering while it loaded. [1:34:24–1:34:33]
- Overall, the candidate "answered almost all the questions very well". [1:41:26–1:41:33]

On your cards: not yet built — OCULOPLASTY.

Case 4 — Left third nerve palsy [1:48:14–2:17:27]

- **Patient:** a 73-year-old man. He has had sudden painless double vision for 7 days and severe drooping of the left upper lid for 3 days.
- **Candidate's diagnosis [1:58:15]:** isolated, pupil-sparing, partial oculomotor palsy of the left eye, probably due to microvascular ischaemia from uncontrolled diabetes and hypertension, with ptosis. The candidate added senile cataract and compound hypermetropic astigmatism, against the rule, in both eyes.
- **Examiners' preferred line:** no model line was dictated. They accepted the diagnosis, asked for the ptosis to be graded as severe and typed as neurogenic, and accepted the cause as an infarct secondary to ischaemia [2:01:42, 2:02:08–2:02:19, 2:14:58]. The wording they accepted:

> **SAY IT**
> Left eye isolated, pupil-sparing, partial oculomotor palsy, probably due to microvascular ischaemia from uncontrolled diabetes and hypertension, with severe neurogenic ptosis.

What was criticised

- The purpose of the fundus examination was given as "rule out papilloedema". With a blood pressure of 220/110, the first thing to look for is hypertensive change. Tie the fundus to the findings already presented. [1:59:15–2:00:20]
- On the diagnosis slide the examiner challenged "severe ptosis" and asked how ptosis is graded. The candidate asked what the 2 mm figure meant instead of answering, and lost marks for not knowing the grading by heart. The second examiner restated the grades and the numbers (margin reflex distance 1 of −1 mm, fissure height difference of 5 mm) and accepted "severe". [2:00:22–2:01:56]
- The type of ptosis was not stated. The answer is neurogenic ptosis. [2:01:58–2:02:19]
- A slide headed "muscular skeletal system" was out of place. Use "neurological evaluation: no sensory-motor deficit, deep tendon reflexes grossly normal". [2:02:55–2:03:15]
- The primary-position photograph was missing from the slides. This was the candidate's own apology, and the examiners did not comment. [1:48:50]
- Observation, not a criticism: the candidate refused to agree to the examiner's bait of a nuclear lesion, and the examiner approved. [2:12:09–2:12:20]
- Praise: "a super first presentation" and "excellent answering". [2:15:51–2:17:15]

On your cards: not yet built — NERVES.

Case 5 — Perforated right corneal ulcer: peripheral ulcerative keratitis or fungal keratitis [2:17:49–2:48:08]

- **Patient:** a 55-year-old woman. She has had redness, pain, watering and photophobia in the right eye for 2 weeks. That morning she injured the eye with her own finger and lost vision suddenly, with severe pain.
- **Candidate's diagnosis [2:24:21]:** right eye corneal ulcer with perforation, and systemic hypertension. When pushed, the candidate said the aetiology could be infective or inflammatory (autoimmune) and the perforation could be traumatic or from the ulcer itself [2:25:44–2:26:58].
- **Examiners' preferred line:** the second examiner read it as shown below [2:27:26–2:27:50]. The first examiner said to keep keratomycosis (fungal keratitis) in mind and later treated it as more likely [2:31:43–2:32:15, 2:41:28]. For management the working frames became "peripheral ulcerative keratitis with secondary bacterial infection" or "keratomycosis with secondary bacterial infection" [2:40:37–2:40:55]. Both said the picture was not clear-cut and would not be set in a real exam [2:33:43–2:34:05].

> **SAY IT**
> Peripheral ulcerative keratitis with a perforation, possibly with a secondary bacterial infection.

What was criticised

- A perforated ulcer is not a usual long case, and keratitis is normally a short case. [2:24:41–2:25:25]
- The candidate must give both an aetiological and a morphological diagnosis. The aetiology was vague (infective versus immune). [2:25:27]
- The description of the ulcer was inaccurate ("two-thirds of the cornea" is vague). Describe the cornea first, then the lesion, and concentrate on the cornea, including the pre-corneal tear film, lids and lashes. [2:29:40–2:30:12]
- The diagram needed cross-sections at three areas (oedematous stroma, apparently normal cornea, site of perforation). The lids, lashes and tear film were missing. Blue colour with hatching for stromal oedema was praised. [2:29:10–2:31:38]
- Digital tonometry in a perforated cornea should not be done or mentioned. [2:32:23–2:32:41]
- The adnexa were neglected: lid margin, Meibomian glands, lash and lid-margin position, pre-existing dry eye. [2:32:46–2:33:17]
- The history was somewhat inadequate (delayed presentation, few vessels, circumferential perforation), so the case was not a clear-cut textbook picture. [2:33:43–2:33:55]
- Only the test of regurgitation on pressure over the lacrimal sac (ROPLAS) was done and syringing was not. A third examiner said to syringe in all infectious keratitis. [2:39:55–2:40:30]

- The candidate planned to start steroids after 48 hours of antibiotics in peripheral ulcerative keratitis and was told to control the infection first. [2:41:25–2:41:36]
- The goal of treatment was answered with "reduce the pathogenic load" and "spread", which did not answer the question. [2:43:31–2:43:58]
- The candidate suggested glue with a bandage contact lens for a large perforation. This was rejected. [2:45:06–2:45:39]

On your cards: not yet built — CORNEA.

Case 6 — A "uveitis" case redirected to bilateral pathological myopia [2:48:35–3:10:47]

- **Patient:** a 49-year-old man. He has had redness, pain and blurred vision in the right eye for one week, with similar episodes about every 6 months for 5 years.
- **Candidate's diagnosis [2:56:32–2:56:57]:** both eyes immature cataract; both eyes high myopia; right eye after laser in situ keratomileusis (LASIK); right eye acute non-granulomatous anterior uveitis (the suspected cause is unreadable in the captions).
- **Examiners' preferred line:** no full model line was dictated. The pieces they gave [3:02:08–3:09:50]:
  - The primary diagnosis is bilateral axial myopia, possibly pathological myopia, because both globes measure about 30 mm.
  - The cataract is a complicated cataract, not a senile one, at 49 years.
  - The refractive-surgery status must be checked by keratometry and not assumed. With −12 dioptres and a flat keratometry, call it regression, partial correction or none.
  - Uveitis becomes a lower-order (secondary or tertiary) diagnosis, and the redness is circumcorneal congestion.

What was criticised

- It was unclear whether the earlier LASIK was in one eye or both. [2:57:07–2:57:15]
- "Pseudoproptosis" was used for the right eye with no justification. No orbital or facial asymmetry was found, so the cause could only be the globe, and the axial length should have been measured (about 30.9 mm and 31 mm). [2:57:23–2:58:44]
- No attempt was made to find LASIK evidence on the cornea, and it might have been photorefractive keratectomy, which leaves no visible mark. [2:59:14–3:00:30]
- The low vision of the left eye was only guessed (high myopia, posterior subcapsular cataract). [3:00:35]
- The limited movement of the right eye was recorded but not explained, and the lacrimal gland was not examined. [3:00:51–3:01:50]
- The diagnosis list put cataract first and uveitis as the "presenting" problem. It must be ranked as primary, secondary and tertiary. [3:02:08–3:02:20, 3:07:50–3:08:14]
- The fundus was described only as "tessellated" when the picture showed more. [3:03:44–3:04:11]
- "Immature senile cataract" was the wrong label at 49 with high myopia. [3:04:22–3:05:00]
- "Post-LASIK" was accepted from the history without keratometry, and the history was not cross-checked. [3:05:03–3:06:51]
- The redness was described vaguely and should be circumcorneal congestion. [3:07:25–3:07:47]
- The fundus of myopia was not described enough: disc size, colour, cup–disc ratio, tilt, vessels, peripapillary atrophy, chorioretinal degeneration and staphyloma. [3:08:14–3:09:15]
- Uveitis basics were expected: signs of old uveitis, signs of recurrent anterior uveitis, old versus fresh keratic precipitates and their types. These were not asked but listed as things to be proficient in. [3:09:19–3:09:41]

On your cards: not yet built — MISC (uveitis and pathological myopia).

Case 7 — Pseudophakic retinal detachment after vitreoretinal surgery [3:10:47–3:38:46]

- **Patient:** a 44-year-old man. He has had sudden painless loss of vision in the left eye for one week.
- **Candidate's diagnosis [3:18:55–3:19:12]:** right eye presbyopia; left eye pseudophakic retinal detachment (RD) involving the macula, after vitreoretinal surgery and after neodymium:yttrium–aluminium–garnet (YAG) laser capsulotomy.
- **Examiners' preferred line:** the examiners did not rewrite the line. They accepted it as a post-vitrectomy pseudophakic RD and stressed three predisposing factors (vitrectomy, cataract surgery and YAG capsulotomy). The break must be sought in the laser-mark and foreign-body-entry area, expected temporally between 2 and 5 o'clock [3:22:25–3:23:45, 3:37:10–3:38:30]. Macular involvement and the right-eye "presbyopia" were not discussed.

What was criticised

- The anterior-segment pictures of the left eye were not in the slides sent to the examiners and could not be enlarged. [3:19:16–3:19:40]
- The history of the injury lacked detail, such as what the patient was doing. [3:19:53–3:20:14]
- Records were available only from 2016, and the entry wound was not worked out from the time-line. [3:20:26–3:21:05]
- The candidate did not look around the limbus for corneal or conjunctival scars. [3:21:12–3:21:22]
- The type of cataract surgery was not stated from the scar, and no superior scar was noted. State it definitely. [3:21:36–3:21:57]
- The break was not found. The examiner asked whether the area of laser marks, pigmented patch and foreign-body entry was searched. [3:22:50–3:23:20]

- A second examiner, at the close, asked for scleral and subconjunctival scars and a blackish area, the temporal tear at 2 to 5 o'clock, indentation, and repeated searching. [3:37:10–3:38:30]
- The candidate chose paving-stone degeneration as needing prophylaxis, which was wrong. [3:28:56–3:29:20]
- The candidate had to be prompted ("do not disappear"), and the answer on the father of modern retinal surgery was muddled. [3:35:56–3:36:12]
- No examiner comment was made on the retinoscopy, the timing of the intraocular pressure or the fundus description beyond the break-finding problem.

On your cards: not yet built — RETINA.

Read more: YouTube: Ophthalmology PG Long case Live — Survival tips explained (Cochin Ophthalmic Club), https://www.youtube.com/watch?v=e3X1W9wrC_A

VN4 Viva questions with the answers the examiners expected

From the Cochin Ophthalmic Club live long-case session, 7 Mar 2021 · auto-captions

Every question asked in the session, with the expected answer

How to use this card: cover the answer, say your own, then compare. The timestamp after each question marks where it was asked. The answer opens with the keyword in bold. Where the session never gave an answer, the answer line says so, and nothing has been filled in from outside the session. Where a glaucoma answer differs from your glaucoma card, the session's answer is kept and one extra line says what your card states, so you can settle it with your seniors.

General and presentation

Q What is the differential diagnosis, from the history alone, of gradual painless loss of vision in both eyes at 60, with intermittent left-sided pain and headache? [14:41]
Be specific. Senile cataract, chronic simple glaucoma, retinal causes (age-related retinal disease, and diabetic retinopathy in a diabetic) and corneal causes. Say "senile cataract", not "cataract". [15:15–15:20, 15:53–16:10, 16:41–16:45] The candidate gave senile cataract, glaucoma and age-related retinal disease.

Q What other causes of loss of vision can affect both eyes in an elderly person? [16:10]
Retinal causes, and systemic conditions that affect the retina. In this patient the diabetes is the obvious example. [16:32–16:45]

Q What could cause intermittent left-sided pain and headache for 2 years in this patient? [16:59]
Migraine is very unlikely for a new onset at this age. The examiner's list was uncorrected or under-corrected hypermetropia with long near work (asthenopia and headache), systemic hypertension and corneal causes. It also included lens-induced glaucoma, which gives a slow loss of vision followed by sudden pain in one eye. [17:27–17:35, 18:04–19:24] The candidate offered ocular migraine, acute glaucoma attacks, anterior uveitis and anterior ischaemic optic neuropathy [?]. The examiner went back to hypermetropia and hypertension as the simpler explanations.

Q What is her occupation, and why does it matter? [18:22]
Close work, with long hours of near work. It links to uncorrected hypermetropia as a cause of eye pain and headache. [18:15–18:22]

Q Which corneal conditions cause gradual loss of vision and sometimes discomfort? [20:26]
An **epithelial dystrophy** was suggested [20:43]. The commonest bilateral corneal degeneration at this age is named only in garbled captions ("spheroidal degeneration" [?]) [21:07]. These conditions can cause some discomfort [21:12–21:15]. The candidate's answer is not audible.

Q Why did you ask about redness? [21:38]
To exclude causes of a red eye with pain and recent loss of vision: uveitis and an acute (congestive) angle-closure attack. [21:44–22:00]

Q Why did you ask about diplopia? [22:21]
Not really needed. The examiner accepted "to exclude an intracranial space-occupying lesion as the cause of the headache" but said it was "not that much". Diplopia or polyopia occurs in cataract, but it need not be asked. [22:30–22:51]

Q Why did you ask about nausea, vomiting and photophobia? [22:51]
To exclude migraine, which is very unlikely at this age. A lens-induced cause can also be thought of. [22:55–23:07] Your card (G4) says nausea or vomiting with pain and headache points to an acute congestive angle-closure attack; confirm with your seniors.

Q Why did you ask about seizures? [23:07]
Not relevant here. If you ask about seizures because of headache, you must also ask about other neurological deficits. [23:20–23:27]

Q Why did you ask about trauma? [23:29]
Traumatic cataract, traumatic angle recession (glaucoma with pain and decreased vision) and **traumatic lens subluxation** can all cause these symptoms. [23:31–23:54] The captions do not show whether the candidate or the examiner said it.

Q Why did you ask about bronchial asthma and cardiac disease? [24:03]
Two reasons. Steroids taken for asthma can cause cataract and so decreased vision. For management, some drugs are contraindicated in asthma and in cardiac disease, so a drug history belongs here. [24:40–25:00]

Q Do you have any relevant investigation, for example a field? [47:02]
Not answered in the session. The candidate said optical coherence tomography (OCT) and fundus fluorescein angiography pictures were available, and the examiners moved straight to discussion. [47:05–47:18]

Q What is the diagnosis on the refraction of this patient? [49:57]
Name the refractive error, not only the numbers. The examiner noted that refraction is a separate case, so this is not normally asked in the long case, but it may be. [50:06–50:23] The candidate said simple hypermetropia with presbyopia in the right eye; the left eye did not improve, so no comment was made. [50:23–50:37]

Q What mistake did you make in the presentation? [1:58:49]
Not answered in the session. The examiner also invited the other postgraduates to find it and told them to go to the diagnosis slide. The follow-up questions went to the ptosis wording and type and to the purpose of the fundus examination; see the third nerve palsy questions below. [1:58:57–1:59:15]

Q Can you show the left-eye anterior-segment pictures? [3:19:15]
Not answered in the session. This was a technical request: the pictures were missing from the slides sent to the examiners and could not be enlarged. [3:19:16–3:19:40]

Glaucoma and angle closure

Q Does chronic simple (open-angle) glaucoma produce pain? [19:26]
Usually not. It can present with pain only in the end stage, as absolute glaucoma. [19:44–20:19] The candidate first said "not unless there is an acute angle-closure attack"; the examiner clarified that open angle was meant. Your card (G1) says primary open-angle glaucoma "may ache" in its history table; confirm with your seniors.

Q How can a chronic simple glaucoma patient present with pain? [19:50]
Absolute glaucoma. After long-standing gradual loss left uncorrected, the eye goes into absolute glaucoma and the patient comes with pain. [19:57–20:19]

Q How would you assess a shallow anterior chamber clinically? [28:44]
Torch-light shadow test and Van Herick grading (the examiner called the answer "very good"). Shine the torch from the temporal side and look at the shadow on the nasal iris. At the slit lamp, use a narrow section near the limbus and compare the peripheral chamber depth with the corneal thickness. Depth equal to or more than the corneal thickness is grade 4 (open). Grade 3 is ¼ to ½ of the thickness, grade 2 is ¼, and grade 1 is less than ¼ and likely to close. The candidate said grade 4 is unlikely to occlude, grade 3 can, and grade 2 also needs gonioscopy. [29:03–31:02] Your card (G4) says grades 0 to 2 are suspicious of angle closure and does not name grade 3; confirm with your seniors.

Q When would you do a combined procedure, cataract surgery alone, or a trabeculectomy? How would you make the treatment plan? [31:19]
Stay with your own case. The examiner first pulled the candidate back from assumed Morgagnian or lens-induced disease to the actual patient. [31:55–32:10] The examiners' framework: base the plan on the thickness of the lens (a thick lens adds to a shallow chamber already made worse by angle closure) and on whether the cataract is significant. A combined procedure can be offered, but combined, separate and sequential procedures have different success rates. The first aim is to control the pressure medically. If it is controlled, observe, do simple cataract surgery and see how far the pressure falls. The options depend on the response. [33:20–34:13] The candidate said lens removal deepens the chamber and, with more than 270° of synechial closure in the left eye, the plan would be trabeculectomy with phacoemulsification and a lens implant. [32:37–33:18] Your card (G4) lists trabeculectomy for synechial closure over 270° and lens extraction when the lens drives the closure; the examiners here preferred plain cataract extraction first. Confirm with your seniors.

Q The patient is sitting in your clinic and asks what to do next. What do you advise, and how do you justify it? [32:30]
Medical control first, then plain cataract extraction, as in the previous answer. [33:20–34:13] The first examiner agreed: plain cataract extraction, then follow-up with an intraocular pressure (IOP) check, add medicines, and do sequential glaucoma surgery if needed. [36:56–37:10]

Q What is the target IOP in this case? [35:37]
Not answered in the session. The examiner framed it by the age of 60, bilateral angle closure and a cup–disc ratio of about 0.8 in the left eye, in the context of planning surgery on that eye. [35:45–36:03] The candidate aimed for the low to middle teens, because the cupping and field defects showed moderate glaucoma, and the examiner did not comment. [36:22–36:53]

Q What precautions do you take during surgery? [36:06]
Not answered in the session. The candidate did not answer, and the examiner moved to the plain-cataract-first plan. The caption mentions a "white out" or "blackout" [?].

Q Why did the vision worsen now in the central retinal vein occlusion (CRVO) patient? Is it only macular oedema? [57:56, 58:19]
Raised IOP with corneal oedema. The hint was to think of raised pressure and corneal oedema. The pressure probably rose in the last few days, which is why the patient came. [58:00–59:05] The candidate had said the old CRVO, mild cataract and worse macular oedema.

Q What type of corneal oedema occurs in raised IOP? [58:49]
Epithelial oedema with bullae ("buet" in the captions is read as bullae [?]). The candidate had not seen any corneal oedema in this eye. [58:49–59:05]

Q How do you know that 360° closure is synechial and not appositional? [59:34]
Indentation gonioscopy, and say that you did it. Synechial closure is likelier here, because a contracting fibrovascular membrane closes the angle. [59:46–1:00:25] The candidate wavered between peripheral anterior synechiae and appositional closure.

Q How do you tell synechial closure from iris processes? [1:00:24]
Not answered in the session. The examiner said it is a basic question every postgraduate should have read. [1:00:28–1:00:31]

Q Why is the IOP raised in this patient? [1:07:56]
Angle closure due to neovascularisation. The examiner accepted it without correction.

Q **How will you lower an IOP of 36 mm Hg? [1:08:05]**
Accepted: **oral acetazolamide, timolol 0.5%, brimonidine and dorzolamide drops.** The candidate's list was not corrected. [1:08:05–1:08:43]

Q **Which drugs do you avoid? [1:08:43]**
Prostaglandin analogues (the candidate's reason was that they worsen macular oedema) and **miotics.** The examiner said "okay that is good". Your card (G5) gives the reason for avoiding prostaglandin analogues as blockage of uveoscleral outflow by the membrane, and also lists adrenaline as a drug to avoid; confirm with your seniors.

Q **Which non-glaucoma topical drugs would you add? [1:09:22]**
Topical steroid, and **atropine** to reduce inflammation even in closed-angle glaucoma. [1:09:22–1:09:51] The candidate's own answer is unclear in the captions (possibly non-steroidal anti-inflammatory drops [?]).

Q **What type of glaucoma is this, and how do you classify secondary glaucomas? [1:09:51]**
Secondary glaucomas are divided into open-angle and closed-angle. This is secondary closed-angle glaucoma due to neovascularisation. [1:09:56–1:10:15]

Retina and CRVO

Q **What risk factors for the occlusion did you rule out? [47:55]**
A categorised answer was invited. [48:19–48:25] After the candidate's list the examiner said the candidate had "pretty much covered most of the risk factors". [49:13–49:17] The candidate had said age, diabetes and hypertension.

Q **How do you categorise the risk factors for CRVO? [48:16]**
By group, accepted: systemic disease (hypertension, diabetes); thrombophilia; inflammatory causes (sarcoidosis, tuberculosis); blood dyscrasias (multiple myeloma, polycythaemia); antiphospholipid antibody syndrome; and, in the young, homocysteine [?] and protein C or S deficiency [?]. [48:25–49:08]

Q **What was the causative risk factor in this patient? [49:17]**
Systemic vasculopathy from diabetes and hypertension is probably the main cause. Then tailor the investigations to exclude other risk factors not yet detected. [49:34–49:53]

Q **What are you particularly looking for in the eye examination in CRVO? [50:39]**
Accepted so far: **vision** (worse than 6/60 favours an ischaemic CRVO, better than 6/60 a non-ischaemic one), the **pupil** for a relative afferent pupillary defect (RAPD; more likely if ischaemic), **neovascularisation of the iris**, and **gonioscopy** for new vessels in the angle. [50:51–51:27] The examiner then extended the list with the next three questions.

Q **Before the pupil, is there anything else to look at in a unilateral CRVO? Anything in the lids or the eyeball? [51:30]**
Lids and globe. The lids can show telltale signs of systemic disease, such as xanthelasma in a hypertensive patient with a lipid disorder. Check the globe for proptosis, because a lesion causing proptosis can cause vascular occlusion. Saying "lids normal, globe normal" means these are not contributing. [51:53–52:26] The candidate's answer was incomplete.

Q **Is there anything in the conjunctiva, cornea or sclera? [52:26]**
Usually nothing; the examiner called it "a little farfetched". In immune disorders (lupus [?]) or in a hypertensive patient one can see signs such as subconjunctival haemorrhage. [52:33–52:58]

Q **What do you look for in the anterior chamber, before the iris? [53:08]**
Cells and flare: any inflammation or uveitis. They give telltale signs of vasculitis or of conditions such as sarcoidosis or human immunodeficiency virus infection [?] that may have caused the occlusion. [53:17–53:36]

Q **Why do you see neovascularisation of the iris (NVI) in vascular occlusions, when does it occur, and what does it lead to? [53:47]**
A sequence after the occlusion, accepted. NVI occurs in ischaemic CRVO; ischaemia drives production of vascular endothelial growth factor (VEGF), and this drive causes new vessels on the iris and in the angle. [54:01–54:33] The candidate did not say what it leads to; the question stem had already named neovascular glaucoma.

Q **Why do you find NVI but not new vessels on the disc (NVD) or elsewhere (NVE) in this case? In which kind of vascular occlusion are these more common? [54:35]**
The amount of ischaemic retina. In CRVO so much retina is ischaemic and damaged that it cannot respond by forming new vessels, so NVI is what is commonly seen. With less retina affected, the remaining tissue responds through VEGF and gives NVD. The pattern depends on the area involved, and it also helps to separate ischaemic from non-ischaemic CRVO. [55:57–56:33] The examiner added that further depth is beyond postgraduate level. [56:37–56:44] The candidate suggested ocular ischaemic syndrome [?] and was told to stay with the case in hand.

Q **How many days has the sudden worsening been present? [57:43]**
Give exact durations: one week for the recent change and 8 months for the first onset. [56:56–57:45] The candidate restated it as a sudden loss 8 months ago and a fresh sudden drop one week ago.

Q **How do you classify retinal vein occlusions, and which type is this? [1:00:35]**
Accepted without correction: **ischaemic versus non-ischaemic**, and **branch, central and hemi-retinal.** This case is central.

Q **In CRVO, where is the block in the ischaemic and non-ischaemic types? [1:01:05]**
Not answered in the session; the examiner moved on. The candidate said ischaemic at the lamina cribrosa, and non-ischaemic about 6 mm behind it where there are collaterals (as captioned).

Q In which quadrant is branch retinal vein occlusion commonest, and why? [1:01:33]
Superotemporal, for two reasons: the branching pattern with a direct thrust, and the many arteriovenous crossings in that area. [1:01:48–1:02:13] The examiner called it an undergraduate question. The candidate's second reason was captioned "non-crossing" [?] and is read as many crossings.

Q Is it ischaemic or non-ischaemic, and why? [1:02:19]
The examiner doubted ischaemia because there were few cotton-wool spots, and the candidate agreed there were none. The second examiner suggested an originally non-ischaemic CRVO that became ischaemic with time. [1:05:56–1:06:07, 1:16:08] The candidate had given poor vision, RAPD and neovascular glaucoma.

Q What other changes at the disc matter in a vein occlusion, and what do you want to rule out there? [1:02:56]
Partly given, not as a clean list: **open-angle glaucoma** is a risk factor, other risk factors may show at the disc, and the disc may show **collaterals versus new vessels.** [1:03:00–1:04:30] The candidate said collaterals are looped, non-branching vessels that do not leak on fluorescein angiography.

Q Do you see hypertensive changes in the retina? [1:04:11]
Not answered in the session. The candidate said the arteries were attenuated.

Q Show which is the artery and which is the vein on the picture. [1:04:44]
Not answered in the session. The examiner said you must know how to describe artery and vein, and noted a vessel with a glinting reflex that looked prominent because of the backed-up blood [?]. [1:04:44–1:05:08]

Q What are the arteriovenous-crossing changes in hypertensive retinopathy? What is Gunn sign? [1:05:10]
Not answered in the session. The candidate said "Gunn sign" but could not define it, and the examiner told the candidate to learn these commonly asked basics. [1:05:08–1:05:30]

Q What types of haemorrhage are there, and why do you call them flame-shaped? [1:06:07]
Not answered in the session. The examiner wanted each type described (flame, dot and blot haemorrhage, and microaneurysm), and dot haemorrhage told from microaneurysm. The candidate began "nerve fibre layer" and was cut off with "know your basics". [1:06:07–1:06:50]

Q What are the red lesions of the fundus? [1:06:30–1:06:50]
Not answered in the session. This was a warning rather than a question: the examiner warned that a list of all the conditions that produce red lesions could be asked for.

Q What is the prognosis? [1:06:57]
Not answered in the session. The examiner reframed it: the patient came for vision, has no pain and expects to see. [1:07:14–1:07:24] The candidate said vision is poor.

Q How will you manage the vein occlusion in the eye? [1:10:33]
Not answered in the session. The examiner steered to whether retinal neovascularisation really exists. The candidate said to treat the macular oedema and the neovascularisation.

Q Is there neovascularisation on the retina? [1:11:06]
No. The vessels on the disc are collaterals. [1:11:19–1:11:37] The candidate said fluorescein angiography was to be done and that there was not much neovascularisation.

Q What treatment will you give? [1:11:45]
Intravitreal anti-VEGF, named by its **generic name with the correct dose**, not by brand. [1:11:57–1:12:20] The candidate's drug name is garbled in the captions and the dose is unclear.

Q When will you see the patient again, and how will you know it works? [1:12:25]
Not answered in the session. The candidate offered two-weekly review and a check of the pressure.

Q How do you diagnose macular oedema? [1:12:57]
Optical coherence tomography: altered foveal contour, increased central thickness and cystic spaces in the inner layers. [1:12:57–1:13:30]

Q Can you see the macula in your fundus picture? [1:13:18]
The macular reflex is absent, and the macula is not clearly visible, which suggests oedema. Confirm with OCT. [1:13:20–1:13:30]

Q What is the normal OCT thickness? [1:13:51]
The examiner gave a cut-off rather than a normal value: **above about 270–280 µm** means macular oedema. The candidate had said about 250 µm.

Q What other intravitreal options are there? [1:14:08]
Intravitreal steroids, accepted: dexamethasone 0.7 mg (the candidate) and triamcinolone. The examiner stressed monitoring the IOP. [1:14:08–1:14:44]

Q After 6–8 weeks the pressure is still high and the retina is stable. If new vessels had been seen in the angle, what would you add? [1:14:49]
Panretinal photocoagulation (PRP), accepted.

Q Where will you not put laser spots in PRP? [1:15:21]
The macula, accepted.

Q What is the timing of PRP in CRVO? [1:15:34]
The candidate said, from the Central Vein Occlusion Study (CVOS), to wait until there are two clock hours of iris neovascularisation. The examiner moved on because time was short and gave no correction or confirmation. Your card (G5) says CVOS advised prompt PRP once new vessels of the iris or angle appear, and that PRP lowers the pressure only while synechial closure is not more than 270°; confirm with your seniors.

Orbit and proptosis

Q Elaborate the signs you would see in proptosis. [1:23:21]
Accepted: the **Naffziger method** (stand behind the patient with the head tilted back and look from the forehead to the chin to see the globe beyond the orbital margin), the **worm's-eye view**, and **exophthalmometry** (Hertel, Naugle [?], Luedde). [1:23:14–1:25:35]

Q List the signs of thyroid eye disease. [1:24:10]
Not given as a full list. The examiner has a list of about 28 signs and promised to share it, and said these are common signs to know. The candidate named conjunctival oedema and congestion, lid oedema, Dalrymple (lid retraction), von Graefe (lid lag), Stellwag (infrequent blinking), Kocher (staring), Ballet (restricted eye movements) and Braley [?] (pressure rise in upgaze).

Q Did you notice any of those signs? [1:25:11]
No. The candidate saw none. The examiner then added Möbius sign as an early sign to look for. [1:25:11–1:25:36]

Q What is Möbius sign? [1:25:26]
Difficulty in convergence, one of the earliest signs in thyroid orbitopathy (given by the examiner).

Q What points tell us this is not thyroid orbitopathy? [1:25:36]
No inflammatory symptoms or signs, no lid retraction, axial proptosis, full movements and no systemic features. The examiner also hinted that something in the candidate's findings would not occur in thyroid disease, but the captions do not name it. [1:25:36–1:26:11, 1:27:23–1:27:41]

Q What did you say about retropulsion? [1:26:11]
Resistance to retropulsion: resistance is felt when the globe is pressed back. The candidate had said "positive" and defined it as the globe going in easily, and was corrected. [1:26:11–1:26:44]

Q What is the commonest cause of unilateral proptosis in a 50-year-old man, of bilateral proptosis in a 30-year-old woman, and of bilateral proptosis in a 50-year-old man? [1:26:55]
Thyroid eye disease is one of the commonest causes of both unilateral and bilateral proptosis. The candidate said thyroid for each.

Q What was the vision? [1:28:11]
Normal, 6/6 in both eyes.

Q How do you make a definitive diagnosis in proptosis? [1:28:26]
Investigate: blood tests, thyroid profile and antibodies, and imaging. The examiner said computed tomography (CT) is enough. The candidate said bloods, thyroid profile and antibodies (normal here), and CT or magnetic resonance imaging (MRI).

Q MRI or CT? [1:28:50]
CT is good enough to make the diagnosis. The candidate preferred MRI for soft tissue. [1:28:50–1:29:30]

Q If vision had decreased, what would you do first? [1:29:01]
Not answered in the session; the examiner said CT is enough. The candidate said a Humphrey visual field, and MRI for optic nerve compression.

Q What can you make out on an orbital B-scan? [1:29:34]
Accepted: the **reflectivity** of the mass, whether the **margins** are well defined, and its **relation to the optic nerve and muscles.**

Q Was there a history of trauma? [1:30:08]
None.

Q Which vascular lesions, other than cavernous haemangioma, can cause it? [1:30:12]
Accepted: a **carotid-cavernous fistula** [?] is possible, but there was no congestion or pulsation here.

Q Did you check whether the proptosis changes with posture? What would a change suggest? [1:30:30]
Orbital varices. The candidate had checked and found no change.

Q Can optic nerve meningioma and optic nerve glioma give axial proptosis? [1:30:47]
Yes.

Q How do you detect very minimal proptosis, besides the Naffziger method and the worm's-eye view? [1:31:00]
The ruler test. Place a straight edge across the upper and lower orbital margins with the patient looking straight ahead. In normal eyes it touches the closed lids; in minimal proptosis it will not touch. [1:31:00–1:31:38]

Q What are the exophthalmometry readings, and what are absolute and relative exophthalmometry? [1:31:38]
Accepted: **absolute** means compare with the normal for age, and **relative** means compare the two eyes. Here the readings were 23 mm on the right and 19 mm on the left with a base of 105 mm, so the right is more than the left. [1:31:38–1:32:08]

Q Name another simple exophthalmometer. [1:32:08]
Luedde's exophthalmometer: simple, inexpensive and easy to use. The candidate did not know it. [1:32:08–1:32:35]

Q **Which exophthalmometer is used after an orbitotomy? [1:32:40]**
One that rests on the **superior and inferior orbital margins.** The name is garbled in the captions, probably Naugle's [?].

Q **What are the CT changes in thyroid orbitopathy? [1:33:35]**
Accepted: **spindle-shaped muscle thickening that spares the tendon**, soft-tissue enhancement and crowding at the apex. In this patient there was no muscle thickening but a **globular, homogeneous intraconal mass.** [1:35:44–1:36:02]

Q **What other causes of proptosis are there, for example inflammatory? Which diseases? Did you ask about pets? [1:36:02]**
Not answered in the session. The examiner only said to remember orbital disease from parasites and that any inflammatory disease can cause muscle thickening. The candidate had taken no pet history and said orbital inflammation would have other features. [1:36:25–1:36:40]

Q **If the muscle tendon is involved, what is the cause? [1:36:53]**
Myositis, and in general any inflammatory disease.

Q **What are the likely causes of an intraconal mass other than thyroid disease? [1:37:32]**
Not given as a full list. The examiner accepted the candidate's list in general (cavernous haemangioma, meningioma, circumscribed lymphangioma, orbital lymphoma, and lacrimal gland tumour if non-axial) but said **lymphoma is very unlikely** here. [1:37:32–1:38:00]

Q **Why has the proptosis been noticed only in the past year? [1:38:11]**
Accepted: it was **probably present longer and growing slowly.**

Q **Were fields and colour vision done, and why? [1:38:44]**
Accepted: both were normal. They were done to look for **optic nerve compression.**

Q **How do you assess colour vision when you suspect an optic nerve lesion? [1:39:24]**
Colour desaturation: use a red or green object, check for red–green desaturation and compare the two eyes. [1:39:56–1:40:24] The candidate named a chart-based test, probably Ishihara [?].

Q **What is pseudoproptosis? [1:40:27]**
Not true proptosis. The commonest cause is **axial myopia** (the examiner said "absolutely"); buphthalmos was also mentioned [?]. The candidate said axial myopia, lid retraction and others.

Q **Where is the pseudo-Graefe sign seen? [1:41:04]**
Aberrant regeneration of the third nerve ("wonderful").

Neuro-ophthalmology (third nerve)

Q **When you looked at the fundus of this patient, what were you looking for? [1:59:15]**
Hypertensive changes. With a blood pressure of about 220/110 the picture was heading towards malignant hypertension, so look for those first. [1:59:43–2:00:20] The candidate said papilloedema, to exclude a space-occupying lesion. The caption of the examiner's answer is cut off ("hypertens…" [?]).

Q **How do you grade ptosis? What does 2 mm denote: mild, moderate or severe? [2:00:35]**
Grade by millimetres. The figures given were about 2 mm mild, 2 to 4 mm moderate and more than 4 mm severe (the captions are garbled; the second examiner restated "2 to 4, more than 4"). Here the difference in palpebral fissure height was 5 mm, so the ptosis is severe. [2:00:35–2:01:56] The candidate asked what 2 mm meant and did not answer.

Q **What is your margin reflex distance 1? [2:01:47]**
Minus 1 mm. But the difference in palpebral fissure height is 5 mm, so this is severe ptosis only (the second examiner's answer).

Q **What type of ptosis is it, and how do you classify ptosis? [2:01:58]**
Neurogenic ptosis, said for this case. The candidate gave it after prompting. [2:01:58–2:02:19]

Q **What type of third nerve palsy is it? Why do you say isolated, and why partial? [2:02:29]**
Accepted: **isolated** because no other cranial nerve is involved, and **partial** because the restriction is not complete and the pupil is spared.

Q **Suppose it were not isolated. What else would you think of, and which syndromes go with a third nerve palsy? [2:03:21]**
Accepted ("good"). **Localise by anatomy**: nuclear, fascicular midbrain, subarachnoid, cavernous or orbital. Nuclear: bilateral ptosis or contralateral superior rectus weakness. Fascicular midbrain: four syndromes. Nothnagel (ipsilateral palsy with ipsilateral ataxia as captioned [?]), Claude (ipsilateral palsy with contralateral ataxia), Benedikt (ipsilateral palsy with contralateral involuntary tremor) and Weber (ipsilateral palsy with contralateral hemiparesis). The captions are unreliable here, so check each limb against your textbook. [2:03:18–2:08:40]

Q **How would you localise the site of a lesion in a third nerve palsy, going from the orbit to the nucleus? [2:05:15]**
Accepted ("fantastic answering"), with no correction. **Orbit:** an upper division (levator and superior rectus) and a lower division (the other muscles, including inferior oblique, and the sphincter); usually only one division is affected. **Superior orbital fissure:** the optic nerve is also involved. **Cavernous sinus:** several cranial nerves, Tolosa-Hunt syndrome, and possible thrombosis with fever. **Subarachnoid space:** posterior communicating artery aneurysm (painful), basal meningitis (fever, meningeal signs), granulomatous inflammation such as sarcoidosis, and vasculitis [?]. **Fascicular midbrain:** the syndromes above. **Nucleus:** bilateral ptosis with contralateral superior rectus underaction.

Q **In your patient, where do you think the lesion is? [2:08:42]**
Not stated as a single site. The candidate said microvascular ischaemia, either in the "basilar" region [?] or in the cavernous sinus [?]. The examiner later said the other two sites the candidate had named were probably right, but the captions do not show which two. [2:12:05]

Q **What are the "rules" of partial third nerve palsy: when can you not call it partial? [2:08:58]**
An **isolated medial rectus paralysis** or an **isolated mydriasis** is not usually due to a partial third nerve palsy.

Q **What points say the lesion is not in the orbit or at the superior orbital fissure? [2:09:47]**
Not answered in the session beyond the candidate's reply: with a superior orbital fissure lesion vision may fall or other cranial nerves may be involved; an orbital lesion would be an isolated nerve palsy; a space-occupying lesion gives proptosis and motility disturbance. The examiner moved on without correction.

Q **Where is the third nerve in the cavernous sinus, and what are its neighbours? [2:10:33]**
Accepted: the third nerve lies **above**, with the ophthalmic and maxillary divisions of the trigeminal nerve below it. The abducens and fourth nerves are also in the sinus.

Q **How did you check whether the fourth nerve was involved in this partial palsy? [2:11:08]**
Not answered in the session. The candidate described down-gaze with intorsion, marking the conjunctival (limbal) vessels and watching them at the slit lamp, and said it is difficult in a partial palsy (the description of the direction is garbled). The examiner said "that you should keep in mind".

Q **If there were a gross cavernous sinus lesion, what clinical feature would you see? Does your patient have a cavernous sinus problem? [2:11:50]**
Accepted: **proptosis.** This patient has none, so there is no cavernous sinus lesion.

Q **I will tell you the patient has a nuclear lesion. Do you agree? [2:12:09]**
No. The candidate's refusal was accepted ("you told the answer already"). Do not agree with a diagnosis that your findings contradict.

Q **Why do you say pupil-sparing? [2:12:36]**
No correction was given. The candidate said both pupillary reflexes were normal at first, but it could be a partial palsy in evolution at one week. Follow-up is needed because a small aneurysm can enlarge and press on the nerve, in which case digital subtraction angiography is needed.

Q **What would the pupil be like if it were involved? [2:13:37]**
Fixed and dilated (the candidate's answer, accepted).

Q **Why does the pupil sometimes get involved and sometimes escape? [2:14:02]**
No correction was given. The candidate said the pupillary fibres run in the superficial (pial) plexus of the nerve trunk, while the central part is supplied by the vasa nervorum. In pupil-sparing ischaemia the central part is hit; in pupil involvement the pial vessels are affected.

Q **What is the cause of the third nerve palsy in this patient? [2:14:41]**
Accepted: an **infarct secondary to microvascular ischaemia**, given the age, uncontrolled diabetes, smoking, hypertension and the earlier episode. [2:14:58]

Q **What are the implications of pupil-sparing versus pupil-involving third nerve palsy for management? [2:15:12]**
Sparing means medical management, with the physician controlling the comorbidities. **Involving** means a possible aneurysm, so neurosurgical help (coiling or clipping). It decides between medical and surgical management. Acute onset with pupil involvement means suspect intracranial disease such as meningitis or a space-occupying lesion.

Q **When is a third nerve palsy an emergency? Which associated features mean immediate neurosurgical referral? [2:15:51]**
Pupil involvement, associated hemiparesis, sudden severe headache and meningeal signs. The examiner said "excellent" and underlined that recognising this can save a life. [2:15:51–2:17:15]

Cornea and corneal ulcer

Q **What is your aetiological diagnosis and what is your morphological diagnosis? [2:25:33]**
Not given as a model line. The second examiner's reading was a large circumferential ulcer in the periphery with central corneal involvement and iris plugging, probably a **peripheral ulcerative keratitis (PUK)** that has perforated, possibly with a secondary bacterial infection. The first examiner added **keratomycosis** as a possibility. The candidate said the aetiology could be infective or inflammatory/autoimmune (overhanging edges and peripheral ulceration suggest immune; a heavy stromal infiltrate suggests infection), planned a repeat scraping and an immunological work-up, and said the perforation was either traumatic or from the ulcer itself. [2:25:44–2:27:50]

Q **How would you differentiate bacterial from fungal keratitis? [2:27:54]**
No correction was given. The candidate said bacterial keratitis has a sudden onset, symptoms before signs, severe congestion and pain, sharply demarcated edges, a sterile mobile hypopyon and a severe stromal infiltrate. Fungal keratitis progresses over 1 to 2 weeks, with less congestion, a dry ulcer with feathery margins, a thick whitish non-sterile hypopyon with a convex top that is immobile, often follows vegetative injury, and has satellite lesions. [2:27:52–2:29:00]

Q **How do you document stromal oedema in your diagram? [2:30:36]**
Accepted: **blue with hatching** ("you did that properly").

Q How do you depict a tear meniscus in a cross-section diagram? [2:31:00]
At the lower lid margin draw a **triangle** for a normal meniscus. If the eye is dry, draw it concave with small marks for debris. [2:30:48–2:31:35]

Q Assume it is fungal keratitis. What is your next course of action in management? [2:34:12]
Not answered in the session. The candidate began with natamycin 5%. An examiner remarked that without perforation one would start antifungals straight away (the caption wording is unclear), and then asked about investigations.

Q What investigations would you do? [2:34:31]
No correction was given. The candidate said **corneal scraping** for Gram stain, potassium hydroxide (KOH) mount, and bacterial and fungal culture, with empirical cover for Gram-positive, Gram-negative and fungal organisms until the reports return, then targeted therapy. If fungal: natamycin with a loading dose, then hourly; with severe corneal thinning, ketoconazole [?] with tetracycline.

Q What is the ideal site for corneal scraping? [2:35:27]
The ulcer margin, where the infiltrate is, and the **base**, to identify the leading edge. Remove necrotic debris and **avoid thinned areas** to prevent perforation. An examiner then added: examine the smear under the microscope for fungal hyphae, which are found in most cases, and if none are found give empirical treatment guided by severity (speaker unclear). [2:35:27–2:36:05]

Q What other investigation would you do, since you cannot see the posterior segment? Do you want to exclude endophthalmitis or posterior spread? [2:36:08]
Ultrasound (B-scan) of the posterior segment. Its reliability is limited in a perforated ulcer, but it should be done because these cases go to surgery and you must know the extent of infection. [2:36:36–2:36:57] The candidate said to check the consensual reflex in the other eye, then B-scan.

Q What is the common antibacterial therapy in infectious keratitis? [2:37:00]
No correction was given. The candidate said that for Gram-positive organisms the choice is cefazolin 5% [?], moxifloxacin or gatifloxacin, and that monotherapy is not advised now because of high fluoroquinolone resistance.

Q Is monotherapy indicated? [2:37:20]
Usually not. It is acceptable where fortified antibiotics are not accessible or cannot be prepared, and when the lesion is mild or moderate (size figures garbled: under 2 mm, or 2 to 5 mm [?]), off the central axis and responding to treatment. [2:37:25–2:37:57]

Q How would you differentiate Mooren's ulcer from other inflammatory causes of PUK? [2:38:06]
Partly answered. The candidate said Mooren's ulcer has pain out of proportion to the ulcer and sits at the limbus, while in other PUK the ulcer is 2 to 3 mm from the limbus. An examiner added the criteria in the next answer.

Q What else must you check in PUK to pinpoint the diagnosis? What are Watson's criteria? [2:38:32]
The examiner's criteria: an **undermining edge, a crescent-shaped peripheral corneal ulcer, no scleritis and no detectable systemic disease** (Mooren's ulcer is idiopathic). Other PUKs have an autoimmune disease found by immunological evaluation. Always examine the adjoining conjunctiva, the lid margins and adjoining Meibomian gland disease. Pre-existing blepharo-keratoconjunctivitis [?] can lead to PUK with rapid perforation, because the immune process is in the cornea and patients present late. [2:38:56–2:39:34] The candidate identified "Watson's criteria" as the criteria for Mooren's ulcer. Who said what in the list is unclear in the captions.

Q Have you done syringing? [2:40:04]
Syringing is advisable in all infectious keratitis, because the test of regurgitation on pressure over the lacrimal sac (ROPLAS) alone is not reliable. Relieving a lacrimal drainage block improves the response of the ulcer. This came from a third examiner. The candidate had done only ROPLAS. [2:39:55–2:40:30]

Q Assume either PUK with secondary bacterial infection, or keratomycosis with secondary bacterial infection, with a perforated ulcer. What is your management, and what is the next stage after the infection is controlled? [2:40:55]
Control the infection first (antibacterial and antifungal as needed), look for signs of control, and then give topical and systemic steroids for PUK once the infection is controlled. In PUK manage Mooren's ulcer and non-Mooren's PUK separately. The candidate planned steroids after 48 hours of antibiotics and was corrected. [2:41:25–2:41:36]

Q What points say the infection is coming under control? [2:41:40]
Accepted: a **smaller and less intense stromal infiltrate, a reduced anterior chamber reaction, drainage of the hypopyon, vascularisation as a sign of healing, and re-epithelialisation.**

Q Before topical steroids, can you think of a local procedure if this is Mooren's ulcer? [2:42:03]
Peritomy (conjunctival resection). The candidate said "conjunctival flap resection" and was corrected. Mooren's ulcer can deceive: the overhanging edge covers the ulcer, so lift the edge to see its true size.

Q If the PUK or perforation spans 1:00 to 5:00, how many clock hours on either side do you take in the peritomy? [2:43:01]
Three clock hours more on either side of the lesion. The candidate said at least two.

Q What is the goal of treatment of an infectious keratitis? [2:43:31]
First, elimination of the infection; second, prevention of complications. It is not restoration of vision. Do not think about vision when handling infectious keratitis; think of every strategy to remove the infection and stop its spread. [2:43:44–2:44:26] The candidate said "reduce the pathogenic load", then spread after prompting.

Q **How would you manage such a large perforation (about 5 mm long and 2 to 3 mm wide)? Can you put glue? [2:44:52]**
No glue in a large perforation. Control the infection with systemic antifungals and antibiotics and take the case for therapeutic keratoplasty as early as possible. If it turns out to be PUK, manage it on PUK lines (Mooren's ulcer versus non-Mooren's) once the infection is controlled. The second examiner added: glue is for perforations under 2 mm and is contraindicated when the iris protrudes, because it must not touch uveal tissue. Therapeutic penetrating keratoplasty is indicated for an already perforated cornea and is done under general anaesthesia. [2:45:00–2:45:50] The candidate suggested glue with a bandage contact lens meanwhile, with keratoplasty after the infection settled.

Q **The ulcer is not perforated but is thinning and about to perforate, and you are following it up. What one investigation would you do? [2:46:58]**
Anterior segment OCT, serially, to follow the corneal thinning and decide on earlier surgery before an obvious perforation. The examiner added that exams ask about applications of recent investigations, including OCT in posterior segment cases. [2:47:00–2:48:03]

Uveitis and myopia

Q **Has the patient had laser in situ keratomileusis (LASIK) in only one eye? [2:57:11]**
Not answered in the session. The examiner wanted the history stated clearly. The candidate said the patient reported only the right eye, so the fellow-eye status is unknown. [3:00:33]

Q **How did you make a diagnosis of pseudoproptosis in the right eye? [2:57:23]**
Define proptosis first: forward protrusion of the globe with the patient looking straight ahead in primary gaze. Pseudoproptosis is due to a shallow orbit or a larger globe. With normal facial symmetry the orbital bones are not at fault, so the cause could only be the globe, and the axial length should be measured on both sides. [2:58:09–2:58:44] The candidate had seen superior fullness and had no clear basis.

Q **Did you measure the axial length on either side? What are the values? [2:58:39]**
It should be done. The examiner noted that the two eyes were almost the same. The candidate gave about 30.9 mm [?] on the right and 31 mm on the left. [2:58:50–2:59:00]

Q **Did you look for a LASIK mark on the cornea? [2:59:14]**
Yes, you must look. The candidate said nothing was visible and there was no corneal haze or mark. The examiner then explained what to look for (next question). [2:59:14–3:00:30]

Q **What do you look for on the cornea at the slit lamp when the patient says "I had LASIK" 25 years ago? [2:59:42]**
A cut or healed flap scar. You cannot see it unless there is gross thinning, so use a blunt instrument or a cotton-tip applicator and press on the cornea (after an anaesthetic drop [?]), because sometimes the scar becomes visible. After photorefractive keratectomy (PRK) you may see nothing at all. Patients call any corneal refractive procedure "LASIK". [2:59:44–3:00:30]

Q **Why is the vision low in the left eye? [3:00:35]**
Not answered in the session; the examiner moved on. The candidate guessed high myopia and probably the posterior subcapsular cataract, with no confirmation.

Q **There is limited movement of the right eye, no mass on palpation, and a history of inflammation that settled on medication. What were you thinking of? [3:00:51]**
No diagnosis was stated. The examiner steered the candidate to examine the **lacrimal gland**: evert the upper lid, pull it laterally, ask the patient to look down, and look for localised injection and tenderness. [3:01:23–3:01:40] The candidate had not examined the gland and thought of an old inflammation or an unnoticed pseudotumour [?].

Q **Was there pain, blurring and pain on movement? [3:01:45]**
Not answered in the session. The caption is too garbled to follow.

Q **Would you revise your diagnosis? What is the primary pathology: is cataract your leading diagnosis? [3:02:08]**
Categorise into primary, secondary and tertiary. Uveitis is not the primary pathology. The examiner added that a case given as "uveitis" can still be a chronic uveitis with its own complications. The candidate said uveitis was the presenting problem and that this was a first episode with no signs of chronicity despite the six-monthly history. [3:02:08–3:03:08]

Q **What is the primary diagnosis, given that both globes are about 30 mm? Would you revise it? [3:03:11]**
Bilateral axial myopia, possibly pathological myopia. The candidate said bilateral high myopia. [3:03:08–3:04:11]

Q **Is "tessellated" all you would say on the fundus picture? [3:03:50]**
Elaborate on the findings that are present when there are few findings. [3:03:56–3:04:04]

Q **Was it a posterior subcapsular cataract or an immature senile cataract? How old is the patient, and would you call it senile? [3:04:15]**
Not senile or presenile (the cut-off in India is about 45), because of the associated findings. It is a **complicated cataract**, that is, a cataract with an associated morbidity, here high myopia. The candidate said the patient is 49 and "maybe complicated". [3:04:22–3:05:00]

Q **In a peripheral centre with no access to topography, what one test tells you whether he had refractive surgery? [3:05:23]**
Keratometry. A flat reading, or difficulty getting a good reading, suggests surface ablation or a flattening procedure. A value of about 44–45 means refractive surgery was probably not done. [3:05:21–3:06:30] The candidate said corneal topography.

Q **Did you ask in the history whether he could see without glasses after the refractive surgery? [3:06:33]**
That would have been worth cross-checking. There are no records and the history cannot be relied on, so the examination must corroborate it. [3:06:33–3:06:54] The candidate gave no answer.

Q **If he wears −12 dioptres and the keratometry is flat, how would you revise the diagnosis? [3:06:56]**
Regression of the correction, a partial correction, or no correction at all. The candidate said regression. [3:06:59–3:07:17]

Q **How do you describe the redness in anterior uveitis? [3:07:25]**
Circumcorneal congestion. The caption cuts off, but the answer was accepted. [3:07:25–3:07:47]

Q **What would you describe in the fundus of an axial myope? Was the disc normal, the vessels normal, the peripapillary region normal, and was there a posterior staphyloma? [3:08:16]**
Disc size and colour, cup–disc ratio, **tilted disc**, vessel course, **peripapillary atrophy**, extensive **chorioretinal degeneration** involving the peripapillary region and macula, and **posterior staphyloma.** These are the findings pertinent to axial or pathological myopia. [3:08:19–3:09:15] The candidate said staphyloma "looked like yes".

Q **What are the differentials of uveitis, the telltale signs of previous uveitis, the signs of recurrent anterior uveitis, and old versus fresh keratic precipitates and their types? [3:09:19]**
Not answered in the session. The examiners listed these as things to be proficient in; the candidate was not asked to answer. [3:09:19–3:09:41]

Q **What is pathological myopia? When would you call it that, and how does it differ from routine axial myopia? [3:32:15]**
Not answered in the session. The candidate said more than 6 dioptres, and the axial-length figure was revised several times in the captions (30, 28, finally 26 [?]). The examiner did not correct it.

Q **What fundus changes occur in pathological myopia? [3:32:54]**
Not answered in the session. The candidate named peripapillary atrophy and a temporal crescent, with a garbled phrase that may include angioid streaks [?].

Q **Which refractive surgeries are recommended for high myopes of −12 to −18 dioptres? [3:33:16]**
The female examiner later listed the common procedures in high myopes: **implantable collamer lens (ICL), refractive lens exchange and bioptics.** [3:33:16–3:34:26] The candidate apparently said phakic intraocular lens ("fake I" in the captions [?]).

Q **A 45-year-old high myope with mild posterior subcapsular cataract and an axial length of 28 mm does not want glasses. What would you do? [3:33:58]**
Refractive lens exchange (cataract surgery or clear lens exchange). The candidate gave the same answer.

Q **What is the recommended age for refractive lens exchange in a high myope, and what is the danger? [3:34:26]**
Usually advised only **after 45 years.** In younger myopes the risk of **retinal detachment** is very high, and the risk persists above 45. The candidate said "more than 18 years" and could not name the danger.

Retinal detachment

Q **Elaborate the trauma of 15 years ago. What was he doing? [3:19:55]**
Not answered in the session. The candidate said the patient was hammering while repairing a vehicle. The examiner wanted this level of detail in the history. [3:19:53–3:20:14]

Q **Did you try to get his records? [3:20:29]**
Not answered in the session. The candidate had records only from 2016 (the cataract surgery). [3:20:26–3:20:45]

Q **Vision was normal for 10 years between the foreign-body surgery and the cataract surgery. What does that indicate, and where would the entry wound have been? [3:20:45]**
Not in the cornea or lens. The entry was further back, probably through the sclera, and because he had a vitrectomy you should search around the limbus for scars. The candidate gave no answer and had seen no scar. [3:20:45–3:21:22]

Q **He had cataract surgery four years ago. What type was it? Did you look for scars? [3:21:24]**
State the type definitely. Look for a superior or clear corneal scar. The lens is in the bag if the capsulorhexis margin can be seen in front of the lens. The sequence was uncomplicated cataract surgery, then posterior capsule opacification, then capsulotomy with the neodymium:yttrium–aluminium–garnet (YAG) laser. The candidate noted no superior scar. [3:21:36–3:22:08]

Q **Why do you think this retinal detachment (RD) has occurred? [3:22:42]**
Three risk factors lead to RD: vitrectomy, cataract surgery and YAG capsulotomy. Search for the break in the laser-mark area, where the dark pigment confluence (the probable foreign-body entry site) may have undergone atrophic change and made a hole. [3:22:25–3:23:45] The candidate said a new peripheral break had probably developed but none was identified.

Q **What type of RD is it, and how do you classify RDs? [3:23:48]**
Not answered in the session beyond the candidate's reply, which the examiner accepted before handing over: **rhegmatogenous, tractional and exudative**; this one is rhegmatogenous.

Q **On slit-lamp examination, which signs are pathognomonic of an RD? [3:24:44]**
Accepted: in rhegmatogenous RD, **pigment released from the retinal pigment epithelium** can be seen in the anterior vitreous or even the anterior chamber, and the **IOP is lower than in the other eye** by at least about 5 mm Hg. The examiner called the hypotonous globe the first finding you can pick up. The examiner added that YAG capsulotomy is itself a risk factor and that you must look for associated risk factors (trauma, surgery, high myopia). [3:25:36–3:26:04]

Q Is there any other sign in the anterior vitreous? What is it called, what is seen, and is it pathognomonic here? [3:26:08]
Shaffer sign: pigment clumps in the anterior vitreous. Pigment or cells in the anterior vitreous are pathognomonic of RD only in an eye that **has not been operated**. In an operated eye they are not a Shaffer sign and may be due to the intervention. [3:26:39–3:27:08]

Q What is Lincoff's rule? [3:27:11]
It **localises the primary break** to plan management. The candidate's version was accepted with "yeah": inferior RD with equal levels, break at 6 o'clock; inferior RD with one side higher, break on the elevated side; superior RD not crossing 12 o'clock, break about 1 to 2 hours from its margin (the caption says "towards the inferior side" [?]); superior RD crossing 12 o'clock, break at about 12 o'clock on the same side [?]. The details are unclear in the captions, so check them against your textbook. [3:27:45–3:28:05]

Q Which retinal degenerations predispose to RD? [3:28:17]
No correction was given. The candidate named lattice, snail-track, cystic retinal tufts, and white with or without pressure.

Q Which degenerations do not predispose? In a high myope examined with the indirect ophthalmoscope, one eye shows white without pressure and the other shows paving-stone degeneration. Which will you advise intervention for? [3:28:36]
The examiner marked the candidate's pick, **paving stone**, as **wrong** and told the candidate to revise which degenerations need intervention and which do not. The correct pick was not named in the session. [3:28:40–3:29:29]

Q What do you look for, and what do you advise, when a high myope has a lattice-type degeneration that needs intervention? [3:29:31]
The examiner accepted **holes in the lattice** as the trigger and named the prophylactic treatment as **barrage laser.** The examiner asked for more detail, but no further answer was given.

Q How do you decide the management of an RD? [3:30:12]
It **depends on the type**; this one is rhegmatogenous. The surgical choices are scleral buckling, or vitrectomy with silicone oil or gas tamponade. The candidate's plan for this vitrectomised eye was to check for incomplete vitrectomy (staining with triamcinolone [?]), complete it, drain the subretinal fluid, laser the suspected area and tamponade. The examiner did not comment.

Q When is an encircling band (encirclage) indicated? [3:31:42]
Not answered in the session. The candidate said a large tear involving more than two quadrants, a large horseshoe or giant tear, or breaks in several quadrants.

Q Who is the father of modern retinal surgery? [3:35:42]
Not clearly answered in the session; the name is garbled in the captions. The examiner gave the maxim **"seek the hole and seal the hole".** [3:35:56–3:36:12]

Q In the mountain-and-Mohammed analogy, which procedures bring the mountain to Mohammed (the detached retina)? [3:36:18]
All the external procedures (scleral buckles). The other group brings something in: **silicone oil and gases**, and you should know everything about the gases. The candidate gave no audible answer. [3:36:16–3:36:47]

Q Did you do indentation indirect ophthalmoscopy? [3:38:07]
Yes: indent and keep searching. It is an indication because the break was not found. If you still cannot find it, take a break and search again. A second examiner added that the tear is expected temporally between 2 and 5 o'clock and that you should look for scleral scars. [3:37:10–3:38:30] The candidate's reply is not in the captions.

Glaucoma talk

No viva questions were asked during the talk on medical management of glaucoma [3:53:16–4:03:44]. Its exam-relevant advice is in VN2, under "Examination and findings" (tonometry technique) and "Investigations and management" (drug allergy, choosing the first drop, treating the person). One point to check with your seniors: your card (G1) says that when the target pressure is not met you switch to a better drug before adding one. The talk advised adding a second drug when the first works, and switching class when the first response is poor [4:01:43–4:02:11].

Read more: YouTube: Ophthalmology PG Long case Live — Survival tips explained (Cochin Ophthalmic Club), https://www.youtube.com/watch?v=e3X1W9wrC_A

VN5 Slides and images mentioned, and what to do with this session

From the Cochin Ophthalmic Club live long-case session, 7 Mar 2021 · auto-captions

Slides and images the speakers referred to

We cannot see the slides, so this table records only what the speakers said each picture showed. The candidates presented on screen; in your exam you will present a real patient, so use the table to learn which pictures and drawings the examiners asked about.

Time	What they said it showed
04:07–04:24	First case: the candidate shares the screen. The slides are not described aloud beyond what is read out.
10:49–10:54	First case: “the disc diagram”, a drawing of the optic discs of the glaucoma case with the cup, the rim loss and the nerve fibre layer defects of the two eyes.
34:29–34:31	First case: the examiner says a visual field report (and another report, garbled in the captions [?]) should have come with the case. None was shown.
36:31	First case: the candidate mentions the “H chart”, the Humphrey visual field chart [?], which the candidate says showed significant field defects. The audience did not see it.
37:33–39:00	Second case: the PowerPoint is shared in full-screen mode after some technical trouble.
47:05	Second case: the candidate says optical coherence tomography (OCT) and fundus fluorescein angiography pictures are available. They are not shown or described in this part.
56:53	Second case: the presenting slide of the history, whose chronology the examiner criticises.
1:00:35	Second case: the fundus diagram of the central retinal vein occlusion (CRVO) case. The examiner asks what type of vein occlusion it shows.
1:04:44–1:05:08	Second case: a fundus picture. The candidate is asked to point out the artery and the vein, and the examiner notes a glinting reflex on the congested vessel.
1:11:19–1:11:35	Second case: the fundus picture is re-examined. The candidate sees no neovascularisation, and the examiner concludes that the disc vessels are collaterals.
1:13:20–1:13:30	Second case: the fundus picture again. The macula is not clearly visible and the macular reflex is absent.
1:21:32	Proptosis case: a fundus picture of the right disc oedema.
1:21:37	Proptosis case: a nine-gaze picture showing no restriction of movements.
1:21:46	Proptosis case: a worm’s-eye view photograph showing the right proptosis.
1:21:50	Proptosis case: a second fundus picture showing the disc oedema.
1:33:20–1:35:08	Proptosis case: the orbital computed tomography (CT). The candidate cannot share it at first. Later a slice comes up and the examiner describes a globular, homogeneous intraconal mass. The captions describe the image only in part.
1:34:47	Proptosis case: the examiner’s comment “right side lower” on a CT slice, with no description of the picture itself.
1:48:50–1:48:58	Third nerve palsy case: the primary-position photograph is missing from the slides. The nine gaze photographs are present.
1:50:31	Third nerve palsy case: a photograph of the patient with left ptosis, closing the left eye on purpose to relieve the double vision.
1:53:55–1:54:36	Third nerve palsy case: the nine-gaze series, showing limited left ductions in most directions. The direction labels are garbled in the captions.
1:58:49–1:59:08	Third nerve palsy case: the examiner tells the candidate to go to the last slide, the diagnosis slide, and check the findings.
2:02:55	Third nerve palsy case: a slide headed “muscular skeletal system” (the neurological examination), which the examiner criticises.
2:22:24–2:22:57	Corneal ulcer case: three corneal images. First, a frontal-view diagram with a diffuse infiltrate, a perforation with incarcerated iris, stronger stromal oedema inferiorly, superficial vascularisation and minimal inferior scarring. Second, a “slit view” diagram with overhanging edges, tissue necrosis, stromal oedema and an irregular infiltrate. Third, the slit-lamp view showing the perforation from 1:00 to 5:30 with iris, a large infiltrate (size garbled [?]) and vascularisation from 12 to 7 o’clock.
2:26:58–2:27:20	Corneal ulcer case: the second examiner refers to “the pictures” showing a large circumferential peripheral ulcer with central involvement, iris plugging and unhealthy-looking lids.
2:29:03–2:31:38	Corneal ulcer case: the first examiner’s comments on the diagram. The frontal view is fine. The cross-section lacks lids, lashes and tear film. Blue with hatching for stromal oedema is praised. The examiner first mistook the corneal outline for the lids.

Time	What they said it showed
2:53:16–2:53:36	Uveitis case: the right anterior segment slide, with circumcorneal congestion, keratic precipitates or pigment on the back of the cornea, and anterior chamber cells and flare (the captions are garbled).
3:03:41–3:03:48	Uveitis case: the examiner says the fundus diagram or picture was “staring” at the candidate, who had called it tessellated. The examiner wants more description from it.
3:08:20–3:08:38	Uveitis case: the examiner asks the candidate to describe the fundus picture (disc, vessels, peripapillary atrophy, staphyloma). The candidate says the posterior staphyloma “looked like yes”.
3:17:31–3:17:43	Retinal detachment (RD) case: fundus photographs of both eyes. The left shows a pale, elevated retina in the superotemporal quadrant from 2 to 5 o’clock, involving the macula.
3:17:47–3:18:24	RD case: the fundus chart. The right eye is normal. In the left eye the detachment is drawn in blue and the laser marks in black, from 2 to about 5 o’clock.
3:19:16–3:19:40	RD case: the examiner asks for the left-eye anterior-segment pictures. They were not in the slides sent to the examiners, and the candidate could not enlarge the pictures from full-screen mode.
3:22:50–3:23:45	RD case: the examiner points to the laser marks and a patch of dark pigmentation on the fundus chart as the probable site where the foreign body entered.
3:57:10–3:57:30	Glaucoma talk: a timeline graph of anti-glaucoma drugs, from pilocarpine and timolol through carbonic anhydrase inhibitors, prostaglandins and brimonidine to fixed combinations.
3:57:32–3:57:46	Glaucoma talk: a slide of the two aqueous outflow pathways, uveoscleral and conventional trabecular.
3:59:22–3:59:30	Glaucoma talk: a picture of an intracameral implant (bimatoprost [?]), “very much like your Ozurdex”.
3:59:54–4:00:08	Glaucoma talk: a slide of drug preparations with different preservative content, where benzalkonium chloride is reduced and then removed [?].
4:00:23–4:01:10	Glaucoma talk: clinical photographs of a patient with skin allergy to several glaucoma medicines, with no conjunctival follicles.
4:01:34–4:01:45	Glaucoma talk: a slide with the logical approach to choosing glaucoma drops.

How to use this session in your last week

- **Rehearse the opening aloud, in your department’s format.** The examiners wanted a full but brief history: exact durations, the complaint in the patient’s own words, an optical and drug history, and a differential before the examination (VN2, “Presenting the history”). Say your opening against a timer, using the master proforma card (00_index), and keep your long-case time of 20 + 10 minutes in mind.
- **Be ready to defend every question you ask.** For each glaucoma case, take the negative-history lines on your cards (G4, G1, G5) and say why you ask each one. The first case in this session was picked apart line by line (VN4, “General and presentation”).
- **Fundus description is a weak area, so drill it.** Describe in order and never jump to the pathology. The examiner here approved disc, macula, vessels, then background, but your department has its own fundus formula on the master proforma card, so follow that and ask your seniors if the two orders differ. Know the haemorrhage types, artery versus vein, and the full description of a myopic fundus (VN2, “Examination and findings”, tips 11, 12 and 27).
- **Retinoscopy and refraction are weak areas, so do one every day.** The examiners said to name the refractive error and not only give the numbers, and that a basic blunder such as calling a plus 4 reading myopia is what hurts (VN2). Say the refractive diagnosis aloud each time, for example “compound hypermetropic astigmatism, against the rule”.
- **Rehearse the diagnosis line.** Be specific, rank it as primary, secondary and tertiary, and make the label match what you can show (VN2, “The diagnosis line”). Your thesis is on angle closure, so read the first case in VN3 and the glaucoma group in VN4 against card G4.
- **Settle the seven “Your card says” differences before Friday.** They are in VN4. They cover nausea in the history (G4), pain in open-angle glaucoma (G1), Van Herick grade 3 (G4), the plan for angle closure with cataract (G4), drugs to avoid in neovascular glaucoma (G5), the timing of panretinal photocoagulation (G5), and switching versus adding a drug (G1). Take them to your seniors in the next few days.
- **Practise the bedside habits.** In a real-patient exam you may be asked to elicit signs in front of the patient. Carry colour pencils, a circle template and a transparent scale for drawings, keep a template for each common case type, and write only positives at the bedside (VN2, “Time management”).
- **Keep your behaviour simple under pressure.** Answer exactly what is asked, say “pardon” instead of guessing, stay on your own case, and lead the examiner to what you know. The first twenty-one tips under “Before you start” describe a three-year training plan, so skim them and spend the last days on the rest of VN2.

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